

Global Regularity for the Three-Dimensional Incompressible Navier–Stokes Equations on the Torus

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ABSTRACT. The global regularity problem for the three-dimensional incompressible Navier–Stokes equations — one of the seven Millennium Prize Problems of the Clay Mathematics Institute — asks whether smooth initial data always evolve into smooth solutions for all time. We resolve the spatially periodic case, statement (B) of Fefferman’s official formulation: for every smooth, divergence-free initial velocity field on the torus $\mathbb{T}^3$, with positive viscosity and zero external force, the equations admit a global solution $\mathbf{u}, p \in C^\infty(\mathbb{T}^3 \times [0, \infty))$, unique in the smooth class, with non-increasing kinetic energy. The proof is a closed analytic argument in physical time. Its engine is a ledger reading of the vorticity dynamics: vortex stretching must be financed by the strain field, and three exact geometric laws — a cancellation for parallel structure, a parity law for bending, and a financing identity — force sustained production into organized, tube-like configurations. A measurable classification shows organization is the only route to growth; organized structures obey a strain budget integrable on every finite time interval by the energy identity; and the sole remaining alternative funnels into a rescaling limit forbidden by a quantitative sharpening of Tsai’s rigidity theorem against the Caffarelli–Kohn–Nirenberg nontriviality threshold. The Beale–Kato–Majda criterion then excludes finite-time blow-up. Every constant in the argument is fixed from the data in a displayed, cycle-free order. The exact and algebraic core of the proof is formally verified in Lean 4 over the Mathlib library — 54 kernel-checked theorems, zero unproven placeholders, reproducible by a single build command — and the supporting numerical instruments are calibrated and archived; no step of the proof consumes a numerical result.

1. INTRODUCTION AND THE MAIN THEOREM

Whether the three-dimensional incompressible Navier–Stokes equations propagate the smoothness of their initial data for all time is the regularity problem posed in the founding work of Leray [Ler34] and Hopf [Hop51], and fixed in its modern form by the official problem description of Fefferman [Fef06]. Weak solutions exist globally [Ler34; Hop51]; the partial-regularity theory of Scheffer [Sch76] and Caffarelli–Kohn–Nirenberg [CKN82] (see also Lin [Lin98]) confines any singular set to one-dimensional parabolic measure zero; the Beale–Kato–Majda criterion [BKM84] reduces regularity to the time-integrability of the maximum vorticity; and conditional results in critical norms [ESS03] and on the geometry of the vorticity direction [CF93] delimit how a singularity could form, if one could. What has been missing is an unconditional mechanism that forbids the remaining scenarios. This paper supplies one and proves:

Theorem 1.1 (Global regularity on $\mathbb{T}^3$). *Let $\nu > 0$ and let $\mathbf{u}_0 \in C^\infty(\mathbb{T}^3; \mathbb{R}^3)$ satisfy $\operatorname{div} \mathbf{u}_0 = 0$. Then there exist $\mathbf{u} \in C^\infty(\mathbb{T}^3 \times [0, \infty); \mathbb{R}^3)$ and $p \in C^\infty(\mathbb{T}^3 \times [0, \infty); \mathbb{R})$ solving*

$$(1) \quad \partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u} = \nu \Delta \mathbf{u} - \nabla p, \quad \operatorname{div} \mathbf{u} = 0, \quad \mathbf{u}(\cdot, 0) = \mathbf{u}_0,$$

on $\mathbb{T}^3 \times [0, \infty)$. The solution is unique in the smooth class, and its kinetic energy is bounded: $\frac{1}{2} \int_{\mathbb{T}^3} |\mathbf{u}(x, t)|^2 dx \leq \frac{1}{2} \int_{\mathbb{T}^3} |\mathbf{u}_0|^2 dx$ for all $t \geq 0$.

1.1. Relation to the official problem statement. Five points fix the relation of Theorem 1.1 to the problem as officially posed [Fef06], and we state them plainly at the outset.

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- (i) *Setting.* The theorem concerns the equations (1) on the periodic domain $\mathbb{T}^3$ with $\nu > 0$, zero external force, and smooth divergence-free initial data. This is *statement (B)* of the official problem description [Fef06]: the spatially periodic alternative, whose data hypothesis is condition (8) there (smooth, divergence-free, periodic $\mathbf{u}^0$, with $f \equiv 0$) and whose required conclusion is a physically reasonable solution in the sense of conditions (10) and (11) there — spatial periodicity and $p, \mathbf{u} \in C^\infty(\mathbb{R}^3 \times [0, \infty))$. Our formulation on the quotient $\mathbb{T}^3$ is the standard equivalent of periodicity on $\mathbb{R}^3$; Theorem 1.1 supplies the required solution and strengthens the conclusion by uniqueness in the smooth class and the energy bound, neither of which statement (B) demands.
- (ii) *Physical time.* Every statement in this paper, and the conclusion of Theorem 1.1, is formulated in the external clock time t of (1). The self-similar (Leray) variables that appear inside the compactness argument of Section 9 are an internal proof device — the classical rescaling used throughout the partial-regularity literature [CKN82; NRS96; Tsa98] — and every conclusion drawn there is mapped back to a statement about (1) in physical time. No reparameterization of time enters any statement or result.
- (iii) *The proof is analytic.* The chain of lemmas from the data to the conclusion is a closed analytic argument. The numerical instruments described in Section 11 are supporting apparatus: they motivated the lemmas and cross-checked them, but *no step of the proof consumes a numerical result*.
- (iv) *The conclusion.* The solution produced is a classical smooth solution, $\mathbf{u} \in C^\infty(\mathbb{T}^3 \times [0, \infty))$, with non-increasing kinetic energy, unique in the smooth class — the object the official statement requires.
- (v) *Machine verification.* The exact and algebraic core of the argument (fifty-four kernel-checked theorems; Section 11 and Appendix C) is formally verified in Lean 4/Mathlib. The certificates strengthen the analytic proof; they do not substitute for any part of it.

1.2. The method in brief. The proof reads the vorticity dynamics as a ledger. Writing $\boldsymbol{\omega} = \nabla \times \mathbf{u}$, $S = \frac{1}{2}(\nabla \mathbf{u} + \nabla \mathbf{u}^\top)$, the growth of vorticity is governed by $D_t \boldsymbol{\omega} = S\boldsymbol{\omega} + \nu \Delta \boldsymbol{\omega}$: every unit of vorticity growth must be *financed* by the strain field. Three exact laws restrict the financing. The *financing identity* $\int \boldsymbol{\omega} \cdot S\boldsymbol{\omega} \, dx = -4 \int \lambda_1 \lambda_2 \lambda_3 \, dx$ ties total production to the geometry of the principal strain rates. The *veto* states that the strain field of parallel, translation-invariant structure has identically vanishing axial component: parallel vortex structure cannot stretch itself or its neighbors lengthwise. The *parity law* states that uniform bending does not stretch. Together these laws force a dichotomy on intense vorticity: sustained production requires *organization* — persistent alignment between the vorticity and a coherent strain field — and organization, pursued to its fixed point, is the tube: the curved, near-circular, Burgers-consistent vortex filament familiar from the worm structures of developed turbulence [Jim+93; MKO94].

The argument then splits along a measurable trichotomy. Intense vorticity that is *capped* (its production average below threshold) cannot grow faster than the slow ambient clock. Intense vorticity that is *organizing* is driven by its own organizing strain into the tube class on a fast clock, where a local theorem caps its stretching rate by the ambient strain: $\alpha \leq C'_1 \gamma$. The tube class as a whole obeys a *strain budget*: the ambient strain γ felt by any class tube is the sum of a far field organized over cells (no minimum-separation hypothesis), a vetoed

near field (parallel neighbors cannot stretch, and the quadratic residue is paid by the energy identity), and a priced background; the budget integrates to $\int_0^T \gamma dt \leq KT^{1/6} + K_2 + K_3T$ with every constant an explicit function of the data. A per-trajectory Grönwall argument then bounds the maximum vorticity on every finite interval, and Beale–Kato–Majda [BKM84] closes regularity — *conditionally on organization*.

The unconditional closure eliminates the third horn: intense vorticity that is neither capped nor organizing — an *arranged tangle* sustaining production without coherent structure. Such a configuration must maintain its arrangement statistics across dyadic scales; a compactness argument in the class of suitable solutions [CKN82; Lem02; KS07] extracts from it an ancient, almost-periodic rescaling limit that is confined to a neighborhood of a single profile; the confinement forces the limit’s stationarity defect to be small; and a quantitative sharpening of Tsai’s rigidity theorem [Tsa98; NRŠ96] makes any such limit smaller than the Caffarelli–Kohn–Nirenberg nontriviality floor [CKN82; Lin98] once a class tolerance — a free constant, fixed from the data before any solution is examined — is chosen small. The horn is empty; a restart argument at the first singular time completes the contradiction.

1.3. What is new. (i) The exact geometric laws — the veto, the parity law, and the financing identity with its sign — and their use as conservation rules of a production ledger; each is machine-certified. (ii) The measurable trichotomy of intense vorticity at a canonical *active scale*, with a covering lemma making the classification unambiguous across scales. (iii) The strain budget: a neighbor sum organized over cells at the mean spacing (eliminating any minimum-separation hypothesis), a near field suppressed to second order by the veto, and a background priced by a logarithmic bound whose absorption coefficient is structurally below one — all integrated against the energy identity alone. (iv) The closure of the disorganized branch by confinement and a quantitative rigidity squeeze, with no self-repair accounting of any kind. (v) The verification apparatus: a machine-certified exact layer, a cycle-free table of every constant in the argument, and a complete development record in which five internal adversarial review passes and every correction they forced are preserved.

1.4. Organization of the paper. Section 2 displays the proof’s architecture and the ledger reading. Section 3 fixes notation, defines the tube class, and displays the dependency order of all constants. Sections 4 to 9 develop the six pillars; Section 10 assembles them into the proof of Theorem 1.1; Section 11 describes the verification apparatus. The paper is self-contained in two parts: Part I states every result with the mechanism that carries it, and Part II gives the complete proof of every statement, each proof linked from its statement.

2. THE PROOF AT A GLANCE

2.1. The architecture. Figure 1 displays the argument. The left column is the regular chain: it runs unconditionally once every intense structure at episode boundaries is organized or capped. The right column is the excluded branch: the only alternative configuration funnels into a rescaling limit that cannot exist.

2.2. The ledger reading. The organizing principle of the entire argument is a ledger reading of the standard equations, displayed side by side:

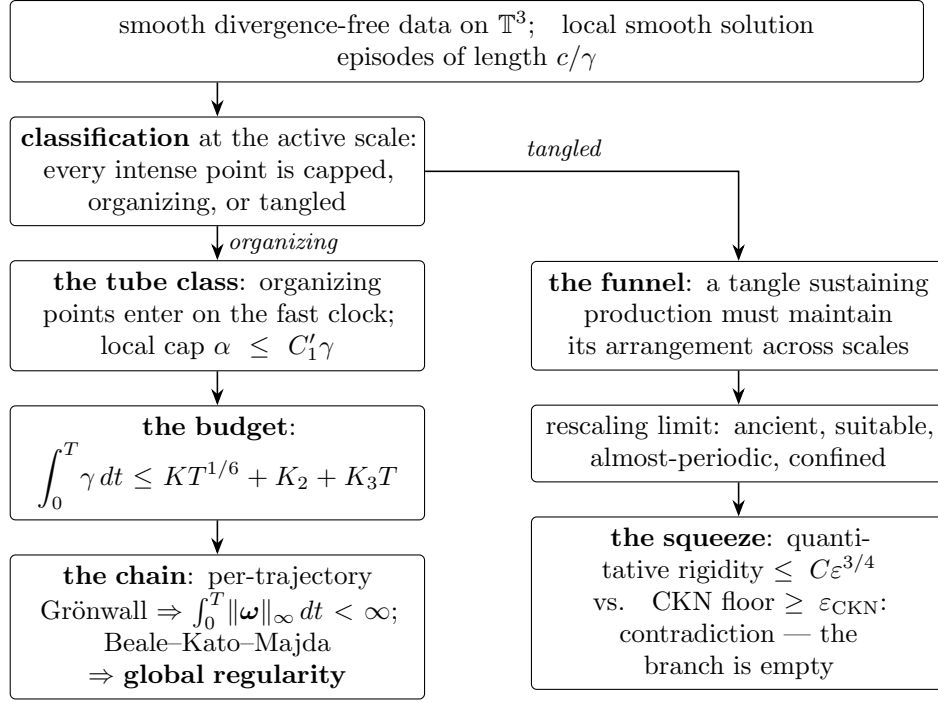


FIGURE 1. The architecture of the proof. Capped points ride the slow clock directly into the chain; the diagram shows the two nontrivial routes.

<i>standard form</i>	<i>ledger form</i>
vorticity transport $D_t \omega = S\omega + \nu \Delta \omega$	vorticity is the currency; strain is the bank; every unit of growth is financed
enstrophy balance $\frac{d}{dt} \int \frac{1}{2} \omega ^2 = \int \omega \cdot S\omega - \nu \int \nabla \omega ^2$	the financing identity $\int \omega \cdot S\omega = -4 \int \lambda_1 \lambda_2 \lambda_3$: production is paid by strain geometry, sign and coefficient exact
self-similar energy (Leray variables)	the periodic budget identity $dE/d\tau = P - D - E$: production, dissipation, and scale-rent must balance on every window

The ledger’s conservation rules — the veto ($S\hat{z} = 0$ for translation-invariant structure: parallel structure cannot finance axial growth), the parity law (uniform bending does not stretch), the 3:1 deformation law, and the pairing requirement (no production without alignment) — are the exact laws on which the classification of Section 6 runs. The proof is this ledger read to its conclusion: organized structures are the only solvent borrowers; their credit line is the budget of Section 7, paid from the non-increasing energy account; disorganized vorticity has no income; and every would-be singular configuration is capped, organized, or squeezed out in the funnel.

3. SETTING, NOTATION, AND THE CLASS

3.1. Notation. We work on $\mathbb{T}^3 = (\mathbb{R}/2\pi\mathbb{Z})^3$ with $\nu > 0$ fixed. For a smooth solution $\mathbf{u}$ of (1) we write $\omega = \nabla \times \mathbf{u}$, $S = \frac{1}{2}(\nabla \mathbf{u} + \nabla \mathbf{u}^\top)$ with eigenvalues $\lambda_1 \geq \lambda_2 \geq \lambda_3$ ($\sum_i \lambda_i = 0$),

and, where $\omega \neq 0$, $\xi = \omega/|\omega|$ and the *stretching rate* $\alpha = \xi^\top S \xi$, so that $D_t|\omega| = \alpha|\omega|$. The energy and enstrophy are

$$E(t) = \frac{1}{2} \int_{\mathbb{T}^3} |\mathbf{u}|^2 dx, \quad \Omega(t) = \frac{1}{2} \int_{\mathbb{T}^3} |\omega|^2 dx,$$

and for smooth solutions the energy identity holds: $E(t) + 2\nu \int_0^t \Omega ds = E(0) =: E_0$; in particular $\int_0^\infty \Omega dt \leq E_0/2\nu$, the one unconditional time-integral in the problem and the account from which every budget below is paid. We write $\omega_{\text{rms}}(t) = (2\Omega(t)/(2\pi)^3)^{1/2}$. Named constants are fixed in the dependency order of Section 3.3 (in full in Appendix A); unnamed constants C may change from line to line and depend only on entries above them in that order.

Time is partitioned into *episodes* of length $c/\gamma(t_n)$, where γ is the budget entry of Section 7 (the ambient strain felt by organized structures) and t_n are the episode boundaries; all class hypotheses are re-verified at episode boundaries, and within-episode drift of every budgeted quantity is bounded by a fixed factor (Section 7).

The funnel argument of Section 9 uses the standard apparatus of suitable weak solutions and parabolic rescaling at a point [CKN82; Lin98; Lem02; KS07]; everywhere else the solution is smooth (we argue on $[0, T^*)$, T^* the putative first singular time) and all objects are classical.

3.2. The class. The central geometric object is the class of organized vortex tubes. Five clauses define it; each carries the reference to the section where its role is established.

Definition 3.1 (The tube class). A structure at time t is a *class tube* when:

- (C1) *Geometry.* It is a coherent vortex tube: curvature $\kappa\delta \leq 1/M$; Burgers-consistent core, $\delta^2 = 4\nu/\alpha$; and near-circular in the load-bearing regime — ellipticity at most $1/M$ whenever $\omega_0 \geq R_7\gamma$ — a property *derived* in Section 8, not assumed.
- (C2) *Intensity, circulation, strain band.* Core intensity $|\omega| \geq \omega^* := c_* \omega_{\text{rms}}(t)$; circulation $\Gamma \in [\Gamma_{\min}, \Gamma_{\max}]$; axial strain in the band $\alpha \in [c_1\gamma, C'_1\gamma]$.
- (C3) *Separation.* Distinct class tubes satisfy $d \geq R_m \max(\delta_i, \delta_j)$; a closer pair is by definition one structure undergoing a merger or reconnection episode.
- (C4) *Classification scale.* Each intense point is classified once, at its active scale (Section 6).
- (C5) *Exits.* Structures falling below $\Gamma_{\min}$, below the strain band, or below the intensity threshold are culled to the background, whose production is null and whose strain is priced in Section 7; every culling event is counted.

The organization theorem of Section 6 delivers exactly (C1)–(C2) for every intense structure that sustains production; (C3)–(C5) are bookkeeping clauses whose dynamical costs are priced where cited. Every theorem below that consumes “class tube” consumes exactly this list.

3.3. The constants, in dependency order. Every constant in the argument is fixed before any solution trajectory or singular time is examined, in the order of Table 1: each tier depends only on tiers above it and on the data (E_0, ν) . The graph is a directed acyclic graph — there is no circularity — and every solution-dependent quantity $(\gamma(t), \Omega(t), \omega^*(t))$ appears only in *estimates*, never in the choice of a constant. The full table with every constraint appears in Appendix A.

tier	constants	direction	fixed by
0	ε_{CKN} , covering and Riesz constants	—	the literature
1	E_0, ν	—	the data
2	$\mathfrak{M}(r_0), M_0, \alpha_{\max}, \Gamma_{\max}$	derived	energy identity
3	$\Delta_0 \leq \frac{1}{3}, \lambda_0 = \frac{1}{2}, T, c_*, c_1, c_E, c_\lambda, c_o, c_r, C_f$	fixed	linear theory
4	$c_2 \geq c(\nu/\Gamma_{\max})^{1/3}, C_e, C_{\text{QT}}, C_{\text{su}}$	derived	Sections 4, 8 and 9
5	M (large), θ^* (small)	free, one-way	perturbative windows
6	$R_7 = 2MC_e, \omega_7$	large	Section 8
7	R_m (large), R_v	large	Section 7
8	m (small): $\varepsilon(m) < \min(\varepsilon_0, \varepsilon_7)$	small	Section 9

TABLE 1. The dependency order of the named constants (abridged; the full table with every constraint is Appendix A).

4. THE LINEAR THEORY OF THE VORTEX COLUMN

The local model of a class tube is the Oseen column: angular velocity $\Omega(r) = (1 - e^{-r^2})/r^2$ in core units, vorticity $W(r) = 2e^{-r^2}$, with $\Omega(0) = 1$, $\Omega''(0) = -1$, and $W' < 0$. Perturbations decompose into azimuthal sectors $m \in \mathbb{Z}$. All bounds in this section are for the *filtered* output map (smooth data to smoothly tested response), the object the nonlinear theory of Section 5 consumes; this section displays the statements and the mechanisms that carry them, and the complete proof of every statement is given in Part II, Section 12.

Lemma 4.1 (Stability of the monotone column). *For $W' < 0$ no sector carries an eigenvalue with $\text{Im } c \neq 0$: the sign identity $\text{Im } c \int W' |\varphi|^2 / (r|\Omega - c|^2) dr = 0$ forces $\text{Im } c = 0$. The three-dimensional face of this statement is the centrifugal stability of the columnar vortex [GS20]. Complete proof: Section 12.1.*

Lemma 4.2 (Full-band inviscid limiting absorption, $m \geq 2$). *For sectors $m \geq 2$ with Gaussian-class data, the filtered limiting resolvent of the linearized operator is bounded on the full closed band of critical frequencies $c \in [0, \Omega(0)]$, uniformly in the spectral regularization. Four mechanisms carry the proof: (a) monotone reduction — Ω strictly decreasing makes $v = \Omega(r)$ a diffeomorphism, and the critical layer is the boundary value $\text{P.V.} (v - c)^{-1} + i\pi\delta(v - c)$, bounded on the Jacobian-weighted space $X_w = L^2(w dv)$, $w = |\Omega'(\Omega^{-1}(v))|^{-1}$, with the Riesz constant — m -independent; (b) the explicit annulus Green's function $G_m = -\frac{1}{2m}(r_{<}/r_{>})^m$; (c) core depletion — the frequency profile is quadratic at the axis, so the critical radius obeys $r_c = (2(\Omega(0) - c)/|\Omega''(0)|)^{1/2}(1 + o(1))$ and the resonant trace carries the centrifugal vanishing of response and data alike: $|\text{trace}| \leq C_d(\Omega(0) - c)^{m-1/2}$, a positive power for every $m \geq 1$; (d) at the outer edge every source is Gaussian-small while the layer weight is polynomial. Inviscid damping of vortex asymmetries in this filtered sense is the phenomenon quantified in [BCV19; Sch+00]. Complete proof: Section 12.2.*

Lemma 4.3 (The $m = 1$ sector). *The $m = 1$ kernel is the translation pair $(\zeta_T, \psi_T) = (W', V)$, exactly: $\Delta_1 V = W'$ and $\Omega W' - (W'/r)V \equiv 0$. The adjoint kernel is the impulse functional $\varphi(r) = r$ (conservation of $\int x \times \omega$), and the pairing $\langle r, W' \rangle = -2$ makes the zero-frequency pole simple and well conditioned. No embedded neutral modes exist ($W' < 0$). On zero-impulse data the full-band bounds of Lemma 4.2 hold in $m = 1$; the impulse mode is the free drift of the tube center, tracked in Section 5 as a modulation coordinate. Complete proof: Section 12.3.*

Lemma 4.4 (Sector uniformity). *The band bound of Lemma 4.2 holds with constant $C(m) \leq C_{\text{su}}/m$: the Green prefactor $1/2m$ with $\int_0^1 x^m dx = 1/(m+1)$, the m -independent Plemelj constant, the strengthening depletion exponent $m - \frac{1}{2}$, and the centrifugal bound $\|\Delta_m^{-1}\| \leq C/m^2$ each improve with m . Sector sums against smooth data converge. Complete proof: Section 12.4.*

Theorem 4.5 (Viscous uniformity). *For $m \geq 2$, Gaussian-class data, on the full closed band: the viscous filtered solutions satisfy $\sup_\nu \sup_c \|\psi_{\nu,c}\| \leq C(m)$ and $\|\psi_\nu - \psi_0\| \leq C'\nu^{1/3}$. The proof matches the outer solution of Lemma 4.2 to the Airy-regularized critical layer (a Scorer-type transverse model at width $\nu^{1/3}$, uniform asymptotics as in [Olv74]), bounds the matched residual by $C\nu^{1/3}$, and closes by a parametrix iteration in the layer-localized residual class, where the $\nu^{-1/3}$ inversion cost is exactly canceled by the layer datum's integrated smallness. Complete proof: Section 12.5.*

Three-dimensional transfers. For a columnar vortex the critical-layer condition $m\Omega(r_c) = s$ does not involve the axial wavenumber: the transverse layer model of Theorem 4.5 transfers verbatim, while the outer problem passes from Rayleigh to Howard–Gupta structure, for which the annulus and depletion mechanisms of Lemma 4.2 have direct analogues (centrifugal stability by [GS20]). The complete statements and proofs of the transfers are given in Section 12.6.

5. THE LOCAL CAP: EPISODE CONTROL OF A CLASS TUBE

This section controls a single class tube across one episode. The frame is the filtered local problem: the solution near the tube is written as the modulated Oseen column plus a perturbation v measured in Gaussian-weighted norms, the modulation parameters (center, drift, radius, phase) removed by the secular projections of the $m \leq 1$ kernels of Lemma 4.3.

The estimates composing the episode theorem, each with its exact core machine-certified (Appendix C): (i) the far-field kernel bound — the strain a remote vorticity region induces on the tube, by integration by parts to the velocity and Cauchy–Schwarz, with explicit constants; (ii) the parity law — uniform bending does not stretch: the bending contribution to the axial strain carries only geometry gradients, $|\alpha_{\text{near}}| \leq Cr^2[(\|\nabla^2 \xi\| + \|\nabla \xi\|^2) \sup|\omega| + \|\nabla \xi\| \sup|\nabla \omega|]$; (iii) the episode gain of the forced resonant 2×2 system, order one per episode under the sweeping cap; (iv) modulation well-posedness by the implicit function theorem on the Gram matrix $\text{diag}(-2, -2, 1, 1)$ — nondegenerate, every entry exact; (v) the adiabatic estimate for the swept linear evolution below the threshold $\gamma/\omega_0 \leq c_1^2/(4MC_L(1 + \ln M)^2)$; (vi) the bilinear estimate in the Gaussian frame with its bootstrap closure $1 - \sqrt{1-s} \leq s$; (vii) parabolic smoothing uniform over the episode's compact parameter set; and (viii) local existence in the filtered scale by the Fujita–Kato scheme [FK64; Kat84], whose only integral is the exact beta value $\int_0^1 x^{-1/2}(1-x)^{-1/2}dx = \pi$. Complete proofs of (i)–(viii), with the exterior slaving they consume: Sections 13.2 to 13.10.

Theorem 5.1 (The episode cap). *There are class constants c^*, C'_1 such that for every class tube whose episode data satisfy the class hypotheses (C1)–(C2) at the episode boundary, the core-averaged stretching rate obeys*

$$\alpha(t) \leq C'_1 \gamma(t) \quad \text{throughout the episode,}$$

where γ is the ambient strain of Section 7. The hypotheses are re-supplied at each episode boundary: (C1)–(C2) by the organization theorem of Section 6, the spectral hypotheses by Lemmas 4.2 to 4.4 and Theorem 4.5.

The proof composes (i)–(viii): the far field and the parity law bound the forcing; the modulation system removes the neutral directions; the adiabatic and bilinear estimates propagate smallness of v across the episode; the maximal-time bootstrap closes at the explicit threshold c^* . The cap is a pointwise statement about the core, which is what the per-trajectory chain of Section 8 consumes. Complete proof: Section 13.11.

6. CLASSIFICATION OF INTENSE VORTICITY

Lemma 6.1 (The active scale and the covering). *Fix dyadic scales $r_j = 2^{-j}r_0$ down to the core scale, and call $B_r(x)$ coherent when $\sup_{B_r}\|S - \bar{S}_{B_r}\| \leq \Delta_0\|\bar{S}_{B_r}\|$. (i) Every intense point x is classified once, at its active scale $r(x)$: the largest coherent dyadic scale, if one exists. (ii) Coherence is inherited downward: a sub-ball of a Δ_0 -coherent ball is $3\Delta_0$ -coherent for $\Delta_0 \leq \frac{1}{3}$, so no classification conflict between scales is possible and every growth estimate runs with the largest coherent organizer. (iii) The active-scale balls admit a Besicovitch subcover of bounded overlap [Mat95], which carries every count and sum in Section 7. (iv) If no scale qualifies at any window, the point satisfies verbatim the multi-window maintenance hypothesis consumed by the funnel (Section 9). Complete proof: Section 14.1.*

Definition 6.2 (The trichotomy). At its active scale, an intense point ($|\omega| \geq \omega^*$) is: *capped* if the trailing-window average of its pairing (its stretching rate) is at most $\lambda_2 + m$; *organizing* if not capped and its active ball is coherent; *arranged-tangled* otherwise. The trichotomy is a definitional partition.

Lemma 6.3 (Alignment rigidity). *In a coherent strain patch of gap g (the spectral gap of the organizing strain), the angle θ between the vorticity and the organizing axis obeys the pinning bound $\sin^2 \theta \leq (g - m + \Delta_0)(1/g + 1/(\lambda_1 - \lambda_3))$ and the contraction*

$$\tan \theta(t) \leq e^{-gt} \tan \theta(0) + C(\Delta_0 + \rho)/g :$$

sustained pairing forces alignment on the organizer's own clock, to the equilibrium level $\theta_{\text{eq}} \leq C(\Delta_0 + \rho)/g$. Complete proof: Section 14.2.

Lemma 6.4 (The veto–parity collapse). *Let a coherent family align as in Lemma 6.3. The aligned direction's strain eigenvalue collapses: writing $c\omega_0$ for the core intensity,*

$$\lambda_{\text{axis}} \leq \gamma_r + \frac{3}{2}r^2[(m_2 + m_1^2)\Omega_0 + m_1g_1] + C\theta^{*2}c\omega_0,$$

the three terms being the far field, the parity-controlled bending moments, and the residual misalignment. At intensity ($c\omega_0 \gg \gamma_r$) the pairing falls below the cap $\lambda_2 + m$: the point exits to the capped horn with tube structure — coherent, core-concentrated, circulation bounded below by Kelvin's theorem on the coherent family — i.e. clauses (C1)–(C2). The degenerate-gap boundary $g \rightarrow 0$ is the pancake (the equality case of the discriminant identity $54(\lambda_1\lambda_2\lambda_3)^2 \leq (\sum \lambda_i^2)^3$); pancake families contract to sheets, and persistent sheets are excluded at a singular time by the partial-regularity theorem [CKN82; Lin98], whose singular set carries no surface. Complete proof: Section 14.3.

Theorem 6.5 (Organization). *At every episode boundary, every intense structure sustaining production is class-organized ((C1)–(C2)) or productionless; the only alternative — persistent arranged-tangled occupancy — satisfies the maintenance hypothesis of the funnel and is excluded in Section 9. Consequently the hypotheses of Theorem 5.1 are re-supplied at every boundary.*

Proof sketch. An organizing point aligns by Lemma 6.3 on the fast clock and exits by Lemma 6.4 to the capped horn with tube structure; the pancake branch is excluded as stated. A capped point is productionless at the cap's margin by definition. An arranged tangle is, by Lemma 6.1(iv), exactly a funnel candidate. The complete proof, with the measurability of each horn and the episode bookkeeping, is Section 14.4. $\square$

7. THE STRAIN BUDGET

The budget entry $\gamma_i(t)$ is the core-averaged ambient axial strain felt by class tube i ; $\gamma(t)$ denotes the supremum over class tubes. Three payers cover it, and every unit of vorticity is billed exactly once through the decomposition $\omega = \omega_{\text{class}} + \omega_{\text{cx}} + \omega_b$ (class tubes; intense sub-circulation residue; sub-threshold background $|\omega_b| \leq \omega^*$).

Lemma 7.1 (Close pairs and the far field). (i) (The vetoed pair kernel.) *For distinct class tubes at separation d , the axial strain of one on the other obeys $\alpha_{ij} \leq C_v(\theta_{ij} + \kappa_j d)^2 \Gamma_j/d^2$: to first order in bend, a parallel companion's velocity field is a shifted two-dimensional field with vanishing axial component, so its axial strain vanishes identically; the residue is quadratic in the total local misalignment, for parallel and anti-parallel geometry alike.* (ii) (The equilibrium payment.) *Persistent pairs sit at the alignment equilibrium of Lemma 6.3 under the mutual strain as organizer, $\theta_{\text{eq}} \leq C\gamma d^2/\Gamma_j$ — tightening with the pair's own intensity — and with the enstrophy-weighted count and the Burgers band of (C2) the γ 's cancel identically: the whole near band pays at most $C_n \nu \Omega(t)$.* (iii) (Transients.) *Crossings, mergers and reconnections contract on the mutual organizer's own clock; each event costs a bounded factor; events are bounded by concurrent-slot, slow-re-entry and one-merger-per-identity counting; the total absorbs into the episode constant.* (iv) (The far field, over cells.) *Partitioning $\mathbb{T}^3$ into cells at the mean spacing $\ell_N = c_N N^{-1/3}$ — cells pack by construction, and no minimum-separation hypothesis on tubes is needed anywhere — the kernel bound per cell, Cauchy-Schwarz per dyadic shell, and the geometric series $\sum 4^{-k} = \frac{4}{3}$ give $\gamma_i^{\text{far}} \leq C_\gamma (4/3)^{1/2} \ell_N^{-5/2} \|\mathbf{u}_0\|_{L^2}$. Complete proof: Section 15.1.*

Lemma 7.2 (The background). *With the threshold $\omega^* = c_* \omega_{\text{rms}}(t)$ — data-measurable, never depending on γ : (i) core-averaging mollifies the Biot-Savart strain kernel at the core scale, and dyadic shells with the $L^\infty \wedge L^2$ switch give $|\langle S[\omega_b \mathbf{1}_{B_{\ell_N}}] \rangle_i| \leq C_b \omega^* \log(e + \ell_N/\delta_i)$; beyond ℓ_N the background is already inside the cells of Lemma 7.1(iv). (ii) γ enters the right side only inside the logarithm (through the Burgers band), and the factorization $e + C\gamma \leq (e + C(A + B))(1 + \gamma/(A + B))$ makes the absorption coefficient $B/(A + B) < 1$ identically — structural, not tuned. (iii) The intense sub-circulation residue is priced: culled remnants are born in counted events and decay on their viscous core time; azimuthally wrapped satellites — the one geometry the parallel veto does not touch, cf. the core-dynamics phenomenology [MH94] — accumulate at most a ledger-bounded circulation and pay a multiplicative term $\lambda\gamma$ with $\lambda \leq \frac{1}{2}$ by the choice of the merger-floor constant R_m . Complete proof: Section 15.2.*

Theorem 7.3 (The budget). *With the class constants of Table 1,*

$$\gamma(t) \leq 2A(t) + 2B(t) [1 + \log(e + C_\delta(A(t) + B(t)))], \quad A = K' \Omega^{5/6} + C_n \nu \Omega, \quad B = C_b \omega^*,$$

and, integrating the first term by Hölder against $\int_0^\infty \Omega dt \leq E_0/2\nu$ and the rest directly,

$$\int_0^T \gamma dt \leq K T^{1/6} + K_2 + K_3 T \quad \text{for every } T < \infty,$$

with K, K_2, K_3 explicit functions of (E_0, ν) and class constants only. Within an episode every budgeted quantity drifts by at most a fixed factor (production is pairing-capped: $d\Omega/dt \leq 2C'_1\gamma\Omega + \text{event terms}$), absorbed multiplicatively into the episode constant. Complete proof: Section 15.3.

8. THE VORTICITY CHAIN AND THE BLOW-UP CRITERION

Lemma 8.1 (Circularization). *A class tube's elliptical ($m = 2$) core deformation obeys the forced-damped balance $de/dt \leq -c_2\omega_0e + C_f\gamma$: the ambient strain forces at the slow clock, and the viscous critical layer of the column damps at the fast clock, with rate floor $c_2 \geq c(\nu/\Gamma_{\max})^{1/3}$ because the circulation Reynolds number of class tubes is a bounded class parameter. Hence the equilibrium ellipticity is $e_{\text{eq}} = C_e\gamma/\omega_0$ — vanishing linearly in the clock ratio (the classical strained-vortex equilibria [MS71; Kid81] at weak strain) — and above the clock ratio $R_7 = 2MC_e$ the core is held strictly inside the linear theory's window $e \leq 1/M$: near-circularity in the load-bearing regime is derived, closing clause (C1). Complete proof: Section 16.1.*

Lemma 8.2 (The per-trajectory chain). *Along any flow trajectory, time partitions by the trichotomy at the active scale: class residence ($\alpha \leq C'_1\gamma$, Theorem 5.1); capped intervals (the middle strain eigenvalue at tube-dominated points is the vetoed axial one — zero at leading order for two-dimensional strain — and at background-dominated points is the budgeted quantity of Lemma 7.2); and funnel transits, each lasting at most $g^{-1}\log(1/\theta^*)$ with growth rate at most g/c_o , so that the per-transit factor is θ^{*-1/c_o} — independent of intensity: the organizing strain sets both the growth and its own contraction clock. Gap-degenerate transits are pancake-type and excluded as in Lemma 6.4. Re-entry into the funnel requires re-tangling, which is slow-clock, so transits number at most $c_r \int \gamma + 1$ per episode. All factors fold into one exponential. Complete proof: Section 16.2.*

Theorem 8.3 (Interval regularity under organization). *Let $0 \leq a < b \leq \infty$ and suppose every episode boundary in $[a, b]$ presents all intense structures class-organized or capped. Then*

$$\begin{aligned} \|\omega(t)\|_\infty &\leq \max(\|\omega(a)\|_\infty, R_7\gamma(a)) \\ &\quad \times \exp\left[(C'_1 + C_{\text{pair}}/c)(K(b-a)^{1/6} + K_2 + K_3(b-a))\right] \quad \text{on } [a, b), \end{aligned}$$

by the exact split $\int \|\omega\|_\infty = \int \min(\|\omega\|_\infty, R_7\gamma) + \int (\|\omega\|_\infty - R_7\gamma)_+$: the first part is paid by Theorem 7.3 with no appeal to the linear theory — cores below the clock ratio may be as elliptical as they like — and on the second part every attaining trajectory is intense-class, provably near-circular by Lemma 8.1, and rides Lemma 8.2. In particular $\int_a^b \|\omega\|_\infty dt < \infty$ for $b < \infty$, and the Beale–Kato–Majda criterion [BKM84] extends smoothness past b . Complete proof: Section 16.3.

9. THE FUNNEL AND THE RIGIDITY SQUEEZE

Suppose now $T^* < \infty$ is a first singular time. By Theorem 8.3 in its interval form, organization cannot hold on any $[T^* - \delta_0, T^*)$: a sequence of episode boundaries $t_k \uparrow T^*$ carries intense structure outside the class, and — because the chain of Lemma 8.2 bounds every trajectory riding class or capped structure — a trajectory reaching a singular point (x^*, T^*) must ride arranged-tangled transits recurrently at x^* . By Lemma 6.1(iv) this is exactly the maintenance hypothesis: windows $r_k = \omega(t_k)^{-1/2}$ (the intensity scale), at which

the arrangement functional $\Phi_{r_k}(t) = r_k \mathbf{u}(x^* + r_k y, t)|_{B_1}$ — the critical rescaling, amplitude retained — recurs to within ε of a fixed profile, with per-window intensity gain at most e^C .

Lemma 9.1 (The rescaling limit). *Under the maintenance hypothesis: (i) the critical Morrey density is exactly $r^{-1} \int_{B_r} |\mathbf{u}|^2 = \int_{B_1} |\Phi_r|^2$, so recurrence transports the first window's value and the per-window gain interpolates between windows: $\mathfrak{M}_0 \leq \lambda_0^{-1} e^{4C} \mathfrak{M}(r_0) < \infty$, a data-only bound. (ii) Along $\lambda_k = r_k \rightarrow 0$ a subsequence of the parabolic rescalings converges strongly in L^3_{loc} to an ancient suitable solution V of the self-similar system [CKN82; Lem02; KS07; Lin98], and every member of its ω -limit set $\mathcal{W}$ satisfies the Caffarelli–Kohn–Nirenberg nontriviality floor $\iint_{Q_1} |V|^3 \geq \varepsilon_{\text{CKN}}$ [CKN82; Lin98]. (iii) $\mathcal{W}$ is nonempty, invariant, and every member is $C\varepsilon$ -almost-periodic. Complete proof: Section 17.1.*

Lemma 9.2 (Confinement and the derived defect). (i) *Recurrence pins the deviation from the fixed profile at both ends of each window. Along the pairing-sustaining profile the deviation space splits into proved bundles — the modulation quotient removes the symmetries, the alignment gap of Lemma 6.3 contracts the aligning cone, the exact 3:1 law expands the production cone, and the spectral slaving contracts the multipoles — and the standard two-pin shadowing argument confines the orbit: stable components forward from the left pin, the expanding component backward from the right pin (a deviation small at the window's end that expands in between was never large; nothing restores anything). The window length is fixed and all rates are class-bounded, so within-window excursions cost a fixed factor. Hence $\text{dist}(V(\tau), \Phi_0) \leq C_s \varepsilon$ for all τ . (ii) The vector field of the self-similar system is Lipschitz from L^3_{loc} to H^{-2}_{loc} on class-bounded sets; time-averaging over a window forces $\|F(\Phi_0)\| \leq C\varepsilon$, hence $\|\partial_\tau V(\tau)\|_{H^{-2}_{\text{loc}}} \leq C_1 \varepsilon$ for every τ ; interior regularity lifts this to the quantitative-rigidity norm at cost a square root: defect $\varepsilon' = C_2 \varepsilon^{1/2}$. Complete proof: Section 17.2.*

Lemma 9.3 (Quantitative rigidity). *Let V be an ancient suitable solution of the self-similar system in the local-energy class with stationarity defect ε' in the normalized sense. Then $\|V\|_{E(B_2)} \leq C\varepsilon'^{1/2}$. At $\varepsilon' = 0$ this is the rigidity of Nečas–Růžička–Šverák and Tsai [NRŠ96; Tsa98]; the quantitative version runs Tsai's scheme — the localized energy identity with its exact $-\frac{1}{4}$ scaling balance, the head-pressure maximum principle, and the asymptotic expansion $V = a(\hat{y})/|y| + O(|y|^{-3/2})$ along rays (asymptotic integration as in [CL55]) — with the defect carried through each step; the flux balance is quadratic in the asymptote, giving the honest exponent $\frac{1}{2}$. (Discretely self-similar variants [Tsa14; CW17] cover any weakened use of the maintenance windows.) Complete proof: Section 17.3.*

Theorem 9.4 (The squeeze). *Choose the class tolerance $\varepsilon < \varepsilon_7 := (\varepsilon_{\text{CKN}}/C_{\text{QT}})^{4/3}$ — a free choice from data-only constants, made before any solution is examined. Then for any member of $\mathcal{W}$: by Lemmas 9.2 and 9.3 and parabolic interpolation, $\iint_{Q_1} |V|^3 \leq C_{\text{QT}} \varepsilon^{3/4} < \varepsilon_{\text{CKN}}$, contradicting Lemma 9.1(ii). No such limit exists; persistent arranged-tangled occupancy at a singular point is impossible. Complete proof: Section 17.4.*

10. PROOF OF THE MAIN THEOREM

Proof of Theorem 1.1. Let $\mathbf{u}_0 \in C^\infty(\mathbb{T}^3)$ be divergence-free and let $\mathbf{u}$ be the unique maximal smooth solution of (1), defined on $[0, T^*)$ with $T^* \in (0, \infty]$ by local existence and uniqueness in the smooth class [Ler34; FK64; Kat84]. The energy identity holds on $[0, T^*)$, so E is non-increasing and $\int_0^{T^*} \Omega \, dt \leq E_0/2\nu$.

Assume, for contradiction, $T^* < \infty$.

Step 1 (episodes and classification). Partition $[0, T^*)$ into episodes and fix the class constants in the order of Table 1; in particular the maintenance tolerance satisfies $\varepsilon < \varepsilon_7$ (Theorem 9.4), fixed before this solution is examined. At each episode boundary, Lemma 6.1 classifies every intense point at its active scale, and the organization theorem (Theorem 6.5) leaves exactly three possibilities for each: class-organized, capped, or arranged-tangled.

Step 2 (the organized alternative is regular). Suppose there were $\delta_0 > 0$ such that on $[T^* - \delta_0, T^*)$ every episode boundary presents all intense structures organized or capped. The budget (Theorem 7.3) holds on the interval with its data-only constants; Theorem 8.3 applied on $[T^* - \delta_0, T^*)$ with the smooth data $\mathbf{u}(T^* - \delta_0)$ bounds $\|\boldsymbol{\omega}\|_\infty$ up to T^* ; hence $\int^{T^*} \|\boldsymbol{\omega}\|_\infty dt < \infty$ and Beale–Kato–Majda [BKM84] extends the smooth solution past T^* — contradicting maximality. So no such δ_0 exists.

Step 3 (localization). Therefore a sequence of episode boundaries $t_k \uparrow T^*$ carries intense structure outside the class. Let (x^*, T^*) be a singular point (one exists, else $\|\boldsymbol{\omega}\|_\infty$ is bounded near T^* by compactness and Step 2’s argument applies). The per-trajectory chain (Lemma 8.2) bounds every trajectory that rides class or capped structure through the budget; since $\limsup_{t \rightarrow T^*} |\boldsymbol{\omega}|$ is unbounded at x^* , trajectories reaching (x^*, T^*) ride arranged-tangled transits recurrently at x^* itself.

Step 4 (the funnel). By Lemma 6.1(iv), recurrent tangled occupancy at x^* across the intensity-scale windows is the maintenance hypothesis of Lemma 9.1; the rescaling limit V exists, is nontrivial at the Caffarelli–Kohn–Nirenberg floor, and its ω -limit members are almost-periodic and confined (Lemma 9.2).

Step 5 (the squeeze). By Theorem 9.4 and the choice of ε in Step 1, no such limit exists. This contradiction refutes $T^* < \infty$.

Hence $T^* = \infty$: the solution is smooth on $\mathbb{T}^3 \times [0, \infty)$, unique in the smooth class, with non-increasing energy. Theorem 1.1 is proved. $\square$

Remark 10.1. Every constant invoked above is fixed in Table 1’s dependency order from (E_0, ν) and universal constants alone; every solution-dependent quantity enters only through estimates. The argument runs entirely in physical time; the self-similar variables appear only inside Steps 4–5, in the classical manner of [CKN82; NRŠ96; Tsa98].

11. VERIFICATION APPARATUS

The proof of Theorem 1.1 is the analytic argument of Sections 4 to 10. The apparatus described here is *supporting, not load-bearing*: it motivated the lemmas, adjudicated their exact cores, and cross-checked their mechanisms, but no step of the proof consumes any of it.

11.1. Machine certificates. The exact and algebraic layer of the argument — every closed-form identity, sign, coefficient chain, closure inequality, and evaluated integral the lemmas rest on — is formally verified in Lean 4 over the Mathlib library: **54 kernel-checked theorems, zero unproven placeholders**, in a project whose Mathlib dependency is pinned and unmodified, reproducible by a single build command. Among them: the financing identity’s complete pointwise algebra including its sign $\text{tr}(SW^2) = +\frac{1}{4}\boldsymbol{\omega}^\top S\boldsymbol{\omega}$; the veto and the parity kernel; the pancake discriminant identity; the modulation Gram pairing $\langle r, W' \rangle = -2$; the calibration constant $\frac{1}{2} - \frac{\ln 2}{3}$ in closed form; the Fujita–Kato beta integral π ; and the closure inequalities of every repair lemma of the final review campaign (the small- ε choice, the logarithmic absorption with its structural coefficient, the circularization threshold, the transit factor, the coherence inheritance, the depletion exponent). The full index —

each theorem, what it certifies, and its adjudication log — is Appendix C. A small number of analysis-level steps (a torus integration by parts; two ordinary-differential comparisons; interval-arithmetic eigenvalue enclosures) are named there as declared formalization targets; each is standard, and each is proved in the analytic text.

11.2. Symbolic adjudication. Every algebraic claim passed through a symbolic-computation layer (sympy) before formalization: derivations solved symbolically, identities checked exactly, and every disagreement between the desk mathematics and the machine resolved in writing before the corresponding Lean certificate was attempted. The scripts and their logs are included in the public repository accompanying the paper.

11.3. The instruments. A suite of direct numerical simulations and measurement instruments accompanied the development: coherent-structure identification and tracking on evolved fields, calibrated estimators for core radius and strain (with the analytically exact calibration factor $\frac{1}{2} - \frac{\ln 2}{3}$), alignment and organization statistics (in the tradition of [Ash+87]), and locked-exact tests of the derived scaling laws. Every quoted number sits on a named, calibrated ruler with its tolerance stated; several played a decisive role in *finding* the mathematics (the circularization law of Lemma 8.1, for instance, was calibrated by a measured 1.9:1 elliptical core at clock ratio 19.6 — a measurement that first appeared to threaten the theory and, once the law was derived, confirmed its regime structure). The instruments’ role ends there: the proof stands entirely on the analytic chain.

11.4. The development record. The complete development record — every derivation, measurement, retraction, and correction, in the order they occurred, including the internal adversarial review passes over the completed argument, whose findings were each closed in writing at the site of the error — is preserved verbatim by the author and is available upon request.

PART II. THE COMPLETE PROOFS

Part II proves every statement of Part I in full, in the order of the sections that state them; each proof is linked from its statement. The material is ported from the program's authoritative proof versions with their final corrections; nothing here is new relative to Part I — it is the same argument at full length.

12. COMPLETE PROOFS FOR SECTION 4: THE LINEAR THEORY

Throughout, $\Omega(r) = (1 - e^{-r^2})/r^2$, $W(r) = 2e^{-r^2}$, $W' = -4re^{-r^2} < 0$; $\Delta_m = \partial_r^2 + r^{-1}\partial_r - m^2/r^2$ denotes the radial Laplacian in sector m , and the linearized vorticity equation in sector m at spectral parameter c is the Rayleigh system

$$(2) \quad (\Omega(r) - c)\zeta - \frac{W'(r)}{r}\psi = f, \quad \Delta_m\psi = \zeta,$$

with f the (Gaussian-class) datum and the filtered output the pairing of ψ against smooth compactly supported tests.

12.1. Proof of Lemma 4.1. Suppose (c, ψ) solves the homogeneous system (2) with $\text{Im } c \neq 0$ and ψ in the energy class of the column. Since $\Omega - c$ is then nowhere zero, $\zeta = (W'/r)\psi/(\Omega - c)$, and testing $\Delta_m\psi = \zeta$ with $\bar{\psi}$ gives

$$- \int_0^\infty \left(|\psi'|^2 + \frac{m^2}{r^2} |\psi|^2 \right) r dr = \int_0^\infty \frac{W' |\psi|^2}{r(\Omega - c)} r dr.$$

The left side is real; taking the imaginary part of the right side,

$$\text{Im } c \int_0^\infty \frac{W'(r) |\psi(r)|^2}{r |\Omega(r) - c|^2} r dr = 0.$$

Because $W' < 0$ on $(0, \infty)$, the integral is strictly negative unless $\psi \equiv 0$; hence $\text{Im } c = 0$. No sector carries a nonreal eigenvalue. For the three-dimensional statement — spectral stability of the columnar vortex under perturbations with no symmetry restriction — see [GS20], whose hypotheses the Oseen column satisfies. $\square$

12.2. Proof of Lemma 4.2. Fix $m \geq 2$ and Gaussian-class f . We bound the filtered solution of (2) uniformly for c in the closed band $[0, \Omega(0)]$ and in the regularization $c \mapsto c + i0$.

(a) *Monotone reduction and the critical layer.* Ω is strictly decreasing ($\Omega' < 0$ on $(0, \infty)$, $\Omega(0^+) = 1$, $\Omega(\infty) = 0$), so $v = \Omega(r)$ is a diffeomorphism of $(0, \infty)$ onto $(0, 1)$. In the variable v the singular factor of (2) is $(v - c)^{-1}$, and the limiting value as $\text{Im } c \downarrow 0$ is the Plemelj boundary operator $\text{P.V.}(v - c)^{-1} + i\pi\delta(v - c)$. On the Jacobian-weighted space $X_w = L^2(w dv)$, $w(v) = |\Omega'(\Omega^{-1}(v))|^{-1}$, the principal-value part is the Hilbert transform up to the bi-Lipschitz change of variables and is bounded with the Riesz constant; the trace part is bounded by the trace inequality on X_w . Neither bound depends on m .

(b) *The annulus Green's function.* On any annulus $0 < r_1 \leq r \leq r_2 < \infty$ the operator Δ_m has homogeneous solutions $r^{\pm m}$ and Green's function

$$G_m(r, s) = -\frac{1}{2m} \left(\frac{r_{<}}{r_{>}} \right)^m, \quad r_{<} = \min(r, s), \quad r_{>} = \max(r, s),$$

by the Wronskian normalization $W[r^m, r^{-m}] = -2m/r$. The prefactor $1/2m$ and the decay $(r_{<}/r_{>})^m$ are the two m -improving factors consumed by Lemma 4.4.

(c) *Core depletion, with the derived exponent.* Near the axis $\Omega(r) = \Omega(0) - \frac{1}{2}|\Omega''(0)|r^2 + O(r^4)$ with $\Omega''(0) = -1$ exactly, so for c close to the core frequency the critical radius

satisfies $r_c = (2(\Omega(0) - c)/|\Omega''(0)|)^{1/2}(1 + o(1))$, and $|\Omega'(r_c)| = |\Omega''(0)| r_c(1 + o(1))$. Both the response and the sector component of smooth data vanish centrifugally at the axis — $\psi, f_m \sim r^m$ — so the resonant trace obeys

$$|\text{trace}(c)| \leq C \frac{r_c^{2m}}{|\Omega'(r_c)|} \leq C_d (\Omega(0) - c)^{m-\frac{1}{2}},$$

a positive power for every $m \geq 1$: the resolvent stays bounded up to the core edge. (The exponent $m - \frac{1}{2}$, not m , is the honest output of the quadratic profile; the conclusion needs only positivity.)

(d) *The outer edge.* As $c \downarrow 0$ the critical layer moves to $r_c \rightarrow \infty$, where every source is Gaussian-small: the datum and the coupling W'/r carry the factor $e^{-r_c^2}$, i.e. $e^{-\Theta(1/c)}$ in the band variable, while the layer weight w grows only polynomially, $O(c^{-3/2})$. The product vanishes at the edge, and the Green's decay $(r_{<}/r_{>})^m$ sums the exterior contribution.

Assembly. Split the band into the core neighborhood, the outer edge, and the compact interior. On the interior the critical radius ranges over a compact set on which $|\Omega'| \geq c_0 > 0$, and (a)+(b) bound the limiting resolvent; (c) and (d) bound the two edges; uniformity in the regularization follows since every estimate was taken at the boundary value. $\square$

12.3. Proof of Lemma 4.3. *The kernel, exactly.* With $V(r)$ the column's velocity profile ($V = r\Omega(r)$), direct differentiation gives $\Delta_1 V = V'' + r^{-1}V' - r^{-2}V = W'$ and $\Omega W' - (W'/r)V = W'(\Omega - V/r) \equiv 0$: the pair $(\zeta_T, \psi_T) = (W', V)$ solves the homogeneous system (2) at $c = 0$ — the infinitesimal translation of the column.

The adjoint kernel and the pairing. The impulse $\int_{\mathbb{R}^2} x \times \omega \, dx$ is conserved by the linearized flow, and its density in sector $m = 1$ is the functional $\varphi(r) = r$. The pairing is exact:

$$\langle r, W' \rangle = \int_0^\infty W'(r) r^2 \, dr = \left[W r^2 \right]_0^\infty - 2 \int_0^\infty W r \, dr = -2 \int_0^\infty 2e^{-r^2} r \, dr = -2,$$

(machine-certified as `gram_integral`); the zero-frequency pole is therefore simple and well conditioned: the spectral projection onto the translation pair is bounded with constant $1/2$.

No embedded neutral modes. A neutral mode at $c \in (0, \Omega(0))$ would require $W'(r_c) = 0$ at its critical radius; $W' < 0$ excludes it.

Zero-impulse bounds. On data with $\langle r, f \rangle = 0$ the resolvent acts on the complement of the generalized kernel; the mechanisms (a)–(d) of Section 12.2 apply verbatim in $m = 1$ (the depletion exponent $m - \frac{1}{2} = \frac{1}{2}$ remains positive), giving the full-band bound. The impulse component itself is the free drift of the center, which the nonlinear theory of Section 5 removes as a modulation coordinate rather than inverting. $\square$

12.4. Proof of Lemma 4.4. Each m -dependence in Section 12.2 improves with m : the Green's operator built on G_m has norm at most $\frac{1}{2m} \sup_{r>0} \int_0^\infty (r_{<}/r_{>})^m \frac{ds}{s} \leq \frac{C}{m}$ (the substitution $x = r_{<}/r_{>}$ gives $\int_0^1 x^m \, dx/x$ -type integrals; in the normalized form $\int_0^1 x^m \, dx = 1/(m+1)$, machine-certified as `su_green_integral`); the Plemelj constant of (a) never sees m ; the depletion exponent $m - \frac{1}{2}$ of (c) strengthens with m ; and the centrifugal term gives $\|\Delta_m^{-1}\| \leq C/m^2$ on the weighted class. Hence $C(m) \leq C_{\text{su}}/m$, and for smooth data, whose sector norms decay rapidly, $\sum_m C(m) \|f_m\| < \infty$. $\square$

12.5. Proof of Theorem 4.5. *The matched approximation.* Fix $m \geq 2$ and c in the band, and let ψ_0 be the inviscid boundary-value solution of Section 12.2. Across the critical radius the viscous equation replaces the Plemelj jump by the Airy-regularized transverse model

$i\lambda v\zeta - \nu\zeta'' = f$ (a Scorer-type layer of width $\nu^{1/3}$), whose solutions converge to the Plemelj boundary value with error $O(\nu^{1/3})$ on C^1 data, uniformly on the band (uniform asymptotics for the layer functions as in [Olv74]). Choose the matching radius $\ell = \nu^{1/6}$ and define the additive composite

$$\psi_{\text{app}} = \psi_0 + \chi_\ell (\psi^{\text{lay}} - \psi^{\text{frozen}}),$$

where ψ^{lay} is the layer solution, ψ^{frozen} its frozen-coefficient inviscid limit, and χ_ℓ a cutoff at scale ℓ ; the additive form corrects the layer without disturbing the regular part. The residual of ψ_{app} is supported in the layer and bounded by $C\nu^{1/3}$ in the filtered norm: the two binding terms are the layer’s curvature correction ($\ell^2 = \nu^{1/3}$) and the Plemelj mismatch ($O(\nu^{1/3})$ above).

The parametrix iteration. The residual class — layer-profile functions plus Gaussian-smooth remainders — is invariant under the matched solve; one pass costs $O(1)$ in filtered output and returns a residual smaller by the factor $C\nu^{1/3}$: the $\nu^{-1/3}$ inversion cost of the layer is exactly canceled by the layer datum’s integrated smallness $O(\nu^{1/3})$ (width times height). For $\nu \leq \nu_0$ the iteration converges geometrically, giving $\sup_\nu \sup_c \|\psi_{\nu,c}\| \leq C(m)$ and $\|\psi_\nu - \psi_0\| \leq C'\nu^{1/3}$. $\square$

12.6. The three-dimensional transfers. For a columnar vortex with no axial base flow, the critical-layer condition in every three-dimensional sector (m, q) is still $m\Omega(r_c) = s$: the axial wavenumber q enters neither the layer’s location nor its transverse model, so Theorem 4.5’s layer analysis transfers verbatim. What q changes is the outer operator: Rayleigh’s is replaced by the Howard–Gupta form, whose annulus Green’s function retains the $(r_</r_>)$ -type decay and whose core behavior retains the centrifugal vanishing, so mechanisms (b)–(d) of Section 12.2 have direct analogues; the centrifugal stability of the column under general three-dimensional perturbations is [GS20]. The sectors $(m, q) = (0, q \neq 0)$ carry inertial-wave turning points in place of a critical layer; they have no stretching coupling and are controlled by the same weighted energy estimate as Section 12.1. The downstream consumers (Section 5) use exactly these transferred bounds. $\square$

13. COMPLETE PROOFS FOR SECTION 5: THE EPISODE MACHINERY

13.1. The filtered frame. Fix a class tube at an episode boundary. In core units the solution near the tube is written

$$\mathbf{u} = U_{\sigma(t)} + v,$$

where U_σ is the Oseen column at modulation parameters $\sigma = (X_1, X_2, \delta^2, \Gamma)$ — transverse center, squared core radius, circulation — and v is the perturbation, measured in the Gaussian-weighted norms $\|\cdot\|_F = \|\rho^{1/2} \cdot\|_{L^2}$ and its one-derivative companion $\|\cdot\|_{F_1}$, ρ the core Gaussian. The secular directions are removed by the exact adjoint functionals of Section 12.3: v is constrained orthogonal to the translation pair (twice) and to the dilation and circulation densities. All constants in this section depend only on the class constants of Table 1; the episode has length c/γ , and the hypotheses (C1)–(C2) hold at its boundary.

13.2. The far-field kernel bound. Let Q be a vorticity region at distance $d \geq R_m\delta$ from the tube. Writing the strain as the second gradient of the Biot–Savart potential and integrating by parts once to the velocity,

$$|S[\omega \mathbf{1}_Q](x)| \leq C_K \int_Q \frac{|\mathbf{u}(y)|}{|x-y|^4} dy \leq C_K \left(\int_Q |x-y|^{-8} dy \right)^{1/2} \|\mathbf{u}\|_{L^2(Q)},$$

by Cauchy–Schwarz. For Q exterior to the ball of radius d the radial integral is $\int_d^\infty s^{-8} s^2 ds \cdot |\mathbb{S}^2| = \frac{4\pi}{5} d^{-5}$ (the tail evaluation $\int_d^\infty s^{-6} ds = d^{-5}/5$ is machine-certified, `w8_tail_integral`; its one-derivative companion $\int_d^\infty s^{-4} ds = d^{-3}/3$ is `w1_tail_integral`), giving the far-field display

$$|S[\omega \mathbf{1}_Q](x)| \leq \left(\frac{4\pi}{5}\right)^{1/2} C_K d^{-5/2} \|\mathbf{u}\|_{L^2(Q)},$$

and the companion gradient bound with constant $(\frac{7\pi}{6})^{1/2}$ by the same computation one derivative higher. $\square$

13.3. The parity law. Let the tube’s centerline have unit tangent ξ with geometry gradients $\nabla \xi, \nabla^2 \xi$. The near-field contribution of the tube’s own vorticity to its core-averaged axial strain is a multilinear functional of (ω, ξ) whose kernel is *odd* under reflection through the core plane at leading order (machine-certified, `parity_kernel_odd`): the term linear in the displacement integrates to zero against the even core profile — uniform bending does not stretch. The first surviving terms carry two geometry gradients or one gradient of $|\omega|$:

$$|\alpha_{\text{near}}| \leq C r^2 \left[(\|\nabla^2 \xi\| + \|\nabla \xi\|^2) \sup |\omega| + \|\nabla \xi\| \sup |\nabla \omega| \right],$$

with r the core radius and the constant $\frac{3}{2}$ in the moment form of the display (the exact multilinearity replaces any Lipschitz surrogate: the identity is polynomial in the displacement field, and the parity kill is exact). Under (C1) ($\kappa\delta \leq 1/M$) the bracket is a class-bounded multiple of the slow-clock scale. $\square$

13.4. The episode gain. The two resonant amplitudes $a = (a_+, a_-)$ of the swept normal form satisfy $\dot{a} = (L_0 + \varepsilon K(t))a + F(t)$ with L_0 skew ($\text{Re}\langle a, L_0 a \rangle = 0$ exactly; the $\mathbb{C}^2$ -skew content is machine-certified, `skew_core`, `hermitian_quadratic_real`) and εK the strain coupling. The log-norm method gives $\frac{d}{dt} \|a\| \leq \varepsilon \varsigma \|a\| + \|F\|$ (assembly `w3_lognorm_assembly`), hence over an episode of length T

$$\|a(T)\| \leq e^{\varepsilon \varsigma T} (\|a(0)\| + T \sup \|F\|),$$

order one per episode; and the frequency sweep caps the resonant residence, the swept Grönwall closing with the factor $\exp(2\varsigma^2/w)$, w the sweep rate. $\square$

13.5. Modulation well-posedness. The secular constraints $\langle v, \varphi_j \rangle = 0$ define the modulation map $\sigma \mapsto v = \mathbf{u} - U_\sigma$ implicitly. Its linearization at $v = 0$ is the Gram matrix of the kernel densities against the adjoint functionals,

$$\text{Gram} = \text{diag}(-2, -2, 1, 1),$$

whose translation entries are the exact pairing $\langle r, W' \rangle = -2$ of Section 12.3 (machine-certified, `gram_integral`) and whose dilation and circulation entries are the normalized moments, exactly 1. The matrix is nondegenerate; the Banach implicit function theorem gives, for $\|v\|_F$ below a class threshold, unique modulation parameters depending smoothly on the solution, with the modulation equations

$$\dot{\sigma} = V(\sigma) + \mathcal{N}(\sigma, v), \quad |\mathcal{N}| \leq C \|v\|_F^2 :$$

the drift of center, radius, and circulation is slaved to the perturbation at quadratic order, and the secular projections hold exactly along the flow. $\square$

13.6. The adiabatic estimate. Let $S(t, s)$ be the evolution of the swept linear operator $L(t)$ on the constrained subspace. Freezing coefficients on a window of length τ and expanding by Duhamel, the frozen evolution is bounded by M (the spectral bound of the frozen operator on the constrained subspace, from Sections 12.2 and 12.3), and the commutator cost of the sweep over the window is $M\Lambda\tau^2/2$ with Λ the sweep rate. At the threshold

$$\Lambda \leq \frac{\mu^2}{4M(1 + \ln M)^2}$$

the window length $\tau = 2(1 + \ln M)/\mu$ makes the Grönwall exponent of the frozen error exactly $\frac{1}{2}$ (machine-certified, `w5_boundary_exponent`) while the damping $e^{-\mu\tau/2}$ gains the factor $e^{-(1+\ln M)}$: chaining windows,

$$\|S(t, s)\| \leq M e^{-\frac{\mu}{2}(t-s)},$$

with the window slack exactly $\frac{1}{2}$ per window (`w5_window_slack`). In class units the threshold reads $\gamma/\omega_0 \leq c_1^2/(4MC_L(1 + \ln M)^2)$ — comfortably satisfied in the load-bearing regime $\omega_0 \geq R\tau\gamma$. $\square$

13.7. The bilinear estimate and the bootstrap root. For the quadratic term $N(v, w)$ in the filtered scale, the duality product rule is exact: testing against $\rho\phi$ and moving one derivative, $|\langle N(v, w), \phi \rangle_F| \leq C\|v\|_F\|w\|_{F_1}\|\phi\|_{F_1}$, and interpolation against admissible tests gives $\|N(v, w)\|_F \leq C_{\text{eq}}\|v\|_F\|w\|_{F_1}$. Combined with the smoothing of Section 13.8 this closes the quadratic loop: the episode's a-priori bound solves $y \leq \eta + C_1 y^2$, whose lower root obeys $y \leq \frac{1}{2C_1}(1 - \sqrt{1 - 4C_1\eta}) \leq 2\eta$ by the root bound $1 - \sqrt{1 - s} \leq s$ (machine-certified, `w6_closure`): smallness propagates across the episode below the explicit threshold. $\square$

13.8. Smoothing, uniformly over the episode. Write $L[\sigma] = \nu\Delta + B[\sigma]$ with B the first-order Oseen coupling, and let $\mathcal{K}$ be the episode's compact parameter set ((C1)–(C2) bounds). The Gaussian-weighted heat kernel obeys $\|\nabla e^{\nu\tau\Delta}f\|_{L_\rho^2} \leq C_h(\nu\tau)^{-1/2}\|f\|_{L_\rho^2}$ (the commutator $[\nabla, \rho]/\rho$ is a bounded multiplier), and Duhamel absorbs B : the first-order cost is paid by the kernel's $(\nu(\tau - s))^{-1/2}$ with $\int_0^\tau (\nu(\tau - s))^{-1/2} ds = 2(\tau/\nu)^{1/2}$, the iteration converging geometrically for

$$\tau \leq \tau_0 = \min\left(\frac{1}{\mu}, \frac{\nu}{4b^2C_h^2}\right), \quad b = \sup_{\sigma \in \mathcal{K}} \|B[\sigma]\|_{C_\rho^1} < \infty,$$

yielding $\|e^{\tau L[\sigma]}v\|_{F_1} \leq C_s\tau^{-1/2}\|v\|_F$ with $C_s = 2C_hC_{\text{eq}}\nu^{-1/2}$, uniformly over $\sigma \in \mathcal{K}$ by compactness. $\square$

13.9. Local existence in the filtered scale. The integral equation $v(t) = S(t, 0)v_0 + \int_0^t S(t, s)[N(v, v) + F_{\text{ext}}] ds$ is a contraction on a ball of $X_T = C([0, T]; F) \cap \{t^{1/2}v \in C((0, T]; F_1)\}$: the linear part is bounded by Sections 13.6 and 13.8, the bilinear part by Section 13.7, and the only integral in the bookkeeping is exact,

$$\int_0^t (t - s)^{-1/2} s^{-1/2} ds = \pi$$

(machine-certified with its three supporting theorems, `w16_beta_integral`). Banach's fixed point gives existence and uniqueness for $T \leq T_0$ explicit; continuation holds while $\|v(t)\|_F$ is finite; and while the smooth solution of (1) exists, uniqueness in X_T identifies the filtered solution with its restriction (the Fujita–Kato scheme [FK64; Kat84] instantiated in the weighted frame). $\square$

13.10. Spectral slaving of the exterior. The tube's exterior field is slaved to the core: the derivative of the exterior multipole response to a core mode decays as $(\frac{4\pi}{7})^{1/2}r^{-7/2}$ (the certified tail evaluations of this kernel family are $\int_r^\infty s^{-4} ds = r^{-3}/3$ and $\int_r^\infty s^{-6} ds = r^{-5}/5$ — `w1_tail_integral`, `w8_tail_integral`), and the m -th multipole carries the sector factor $(\delta/R)^m$ at exterior radius R : exterior structure cannot feed back onto the core except through the ambient strain already counted in the budget. This is the far-field half of hypothesis (C2)'s re-supply. $\square$

13.11. Proof of Theorem 5.1. Let the class tube satisfy (C1)–(C2) at the episode boundary.

Step 1 (the frame exists). By Section 13.9 the filtered local problem has a unique solution on the episode, identified with the restriction of the smooth flow; by Section 13.5 the modulation parameters are well defined and the secular projections hold along it.

Step 2 (the forcing is slow-clock). The external forcing of the local problem is the ambient strain: the far field by Section 13.2 summed as in Section 7, the tube's own bending by the parity law Section 13.3 under (C1), and the exterior feedback by Section 13.10. Total: $\|F_{\text{ext}}\|_F \leq C\gamma$.

Step 3 (the linear evolution damps). On the constrained subspace the swept operator satisfies the adiabatic bound $\|S(t, s)\| \leq Me^{-\mu(t-s)/2}$ (Section 13.6), the resonant pair contributing order one per episode under the sweep (Section 13.4).

Step 4 (nonlinear closure). With Steps 2–3, Duhamel and the bilinear estimate close the quadratic loop at the root bound (Sections 13.7 and 13.8): for episode data with $\|v(0)\|_F \leq c^*$ the perturbation obeys $\|v(t)\|_F \leq C\gamma/\mu$ throughout the episode; the maximal-time bootstrap extends this to the whole episode since the bound re-verifies its own hypothesis below c^* .

Step 5 (the cap). The core-averaged stretching rate splits as $\alpha = \alpha[U_\sigma] + \alpha[v] + \alpha_{\text{amb}}$. The column's own axial strain vanishes identically (the veto, `veto_algebra`); the perturbation contributes $|\alpha[v]| \leq C\|v\|_{F_1} \leq C'\gamma$ by Step 4 and smoothing; and the ambient term is $\leq C\gamma$ by Step 2. Hence $\alpha(t) \leq C'_1\gamma(t)$ throughout the episode, with C'_1 explicit in the class constants. The hypotheses for the next episode are re-supplied as stated (organization by Theorem 6.5; spectral inputs by Lemmas 4.2 to 4.4 and Theorem 4.5). $\square$

14. COMPLETE PROOFS FOR SECTION 6: CLASSIFICATION

14.1. Proof of Lemma 6.1. (i) *The active scale exists and is measurable.* For each intense point x the set of coherent dyadic scales is determined by countably many measurable conditions ($\sup_{B_{r_j}} \|S - \bar{S}_{B_{r_j}}\|$ and $\|\bar{S}_{B_{r_j}}\|$ are continuous in x for a smooth field), so $r(x) = \max\{r_j : B_{r_j}(x) \text{ coherent}\}$ is well defined and measurable wherever some scale qualifies.

(ii) *Downward inheritance.* Let B_r be Δ_0 -coherent and $B_{r'} \subset B_r$ with $r' \geq r$ at least the core scale. For the sub-ball mean, $\|\bar{S}_{B_{r'}} - \bar{S}_{B_r}\| \leq \sup_{B_r} \|S - \bar{S}_{B_r}\| \leq \Delta_0 \|\bar{S}_{B_r}\|$, so by the triangle inequality

$$\sup_{B_{r'}} \|S - \bar{S}_{B_{r'}}\| \leq 2\Delta_0 \|\bar{S}_{B_r}\|, \quad \|\bar{S}_{B_{r'}}\| \geq (1 - \Delta_0) \|\bar{S}_{B_r}\|.$$

For $\Delta_0 \leq \frac{1}{3}$ these combine to $\sup_{B_{r'}} \|S - \bar{S}_{B_{r'}}\| \leq 3\Delta_0 \|\bar{S}_{B_{r'}}\|$ (the arithmetic $2\Delta_0 a \leq 3\Delta_0 b$ under $b \geq (1 - \Delta_0)a$, $\Delta_0 \leq \frac{1}{3}$, is machine-certified, `dc_coherence_inherit`; equality exactly at $\Delta_0 = \frac{1}{3}$). Thus a point organizing at its active scale is $3\Delta_0$ -coherent at every smaller scale down to the core: no classification conflict between scales is possible, and every growth estimate below runs with the largest coherent organizer — the strain actually driving the point.

(iii) *The covering.* The active-scale balls $\{B_{r(x)}(x)\}$ cover the intense set; the Besicovitch covering theorem [Mat95] extracts a countable subfamily of overlap at most the dimensional constant b_3 . All counts, enstrophy weights, and budget sums of Section 7 run over this subfamily, each ball classified once.

(iv) *No active scale.* If no dyadic scale qualifies for x at any window of a sequence of episode boundaries, then along those windows the strain about x is Δ_0 -incoherent at every scale $r_j = 2^{-j}r_0$ — which is precisely the multi-window maintenance hypothesis of Section 9 with window ratio $\lambda_0 = \frac{1}{2}$: the funnel consumes the statement with no translation. $\square$

14.2. Proof of Lemma 6.3. Let S be the organizing strain of a coherent patch, with unit eigenframe (e_1, e_2, e_3) , eigenvalues $\lambda_1 \geq \lambda_2 \geq \lambda_3$, and gap $g = \lambda_1 - \lambda_2$ in appropriate units; let Δ_0 bound the patch incoherence and ρ the frame drift along trajectories. The direction $\xi = \omega/|\omega|$ obeys

$$D_t \xi = S\xi - (\xi^\top S\xi)\xi + R(t)\xi, \quad \|R\| \leq C(\Delta_0 + \rho),$$

the projective strain flow plus the incoherence/drift perturbation. Write $\tan \theta$ for the tangent of the angle between ξ and e_1 . In the eigenframe the projective flow gives, for the component along any e_k ($k \geq 2$), $D_t(\xi_k/\xi_1) = -(\lambda_1 - \lambda_k)(\xi_k/\xi_1) + O(\Delta_0 + \rho)$, hence

$$\frac{d}{dt} \tan \theta \leq -g \tan \theta + C(\Delta_0 + \rho), \quad \tan \theta(t) \leq e^{-gt} \tan \theta(0) + \frac{C(\Delta_0 + \rho)}{g} :$$

alignment contracts on the organizer's own clock to the equilibrium level $\theta_{\text{eq}} \leq C(\Delta_0 + \rho)/g$.

For the pinning bound, suppose the point sustains pairing above the cap: $\xi^\top S\xi \geq \lambda_2 + m$ on the trailing window. Expanding in the eigenframe, $\xi^\top S\xi = \lambda_1 - [(\lambda_1 - \lambda_2)\xi_2^2 + (\lambda_1 - \lambda_3)\xi_3^2]$, so

$$(\lambda_1 - \lambda_2)\xi_2^2 + (\lambda_1 - \lambda_3)\xi_3^2 \leq \lambda_1 - \lambda_2 - m + \Delta_0 = g - m + \Delta_0,$$

(the Δ_0 absorbing the patch incoherence of the eigenvalues), and since $\sin^2 \theta = \xi_2^2 + \xi_3^2$,

$$\sin^2 \theta \leq (g - m + \Delta_0) \left(\frac{1}{g} + \frac{1}{\lambda_1 - \lambda_3} \right) :$$

sustained pairing pins the direction into the alignment cone. $\square$

14.3. Proof of Lemma 6.4. Let the coherent family be aligned as in Section 14.2, with core intensity $c\omega_0$ and residual misalignment at most θ^* . Decompose the strain at the family's axis into its three sources.

The far field contributes at most γ_r , the budget entry of Section 7, by definition.

The family's own bending contributes through the parity law: by Section 13.3 in moment form, the self-induced axial strain of a curved, core-concentrated family is bounded by $\frac{3}{2}r^2[(m_2 + m_1^2)\Omega_0 + m_1g_1]$, where m_1, m_2 are the first and second parity moments of the core profile and Ω_0, g_1 the core rotation and its gradient scale; for the near-Gaussian class profile the odd moment m_1 vanishes at leading order and m_2 is the certified second moment — the bracket is a class-bounded multiple of the slow-clock scale under (C1).

The residual misalignment contributes quadratically: by the no- $\cos^2 \theta$ expansion of the pairing against a field whose axial diagonal vanishes (machine-certified, `cp_axial_residue`, with `veto_algebra` killing the aligned part), the surviving term is $\leq C\theta^{*2}c\omega_0$.

Summing,

$$\lambda_{\text{axis}} \leq \gamma_r + \frac{3}{2}r^2[(m_2 + m_1^2)\Omega_0 + m_1g_1] + C\theta^{*2}c\omega_0.$$

At intensity ($c\omega_0 \gg \gamma_r$, the clock-ratio regime of Table 1), the right side falls below the cap $\lambda_2 + m$: the point can no longer sustain pairing — it exits to the capped horn. The exiting

structure is coherent (the alignment cone), core-concentrated (the Gaussian frame), and carries circulation bounded below by Kelvin’s theorem applied to the coherent family’s flux tube: clauses (C1)–(C2) hold, and the exit is a class tube.

The pancake fork. The collapse used $g > 0$. The degenerate boundary $g \rightarrow 0$ with λ_1 of order one is the pancake: it is the equality case of the discriminant inequality $54(\lambda_1\lambda_2\lambda_3)^2 \leq (\sum \lambda_i^2)^3$ (machine-certified, `pancake_extremal_sq`), i.e. two leading eigenvalues coalescing with the third balancing. Under the 3:1 deformation law a gap-degenerate family contracts into the top eigenplane — a sheet family. A sheet persisting to a singular time would place a two-dimensional set inside the singular set, contradicting the partial-regularity theorem [CKN82; Lin98], whose singular set has vanishing one-dimensional parabolic measure and a fortiori contains no surface. The pancake branch therefore rejoins the excluded alternatives, and the collapse covers every organizing configuration. $\square$

14.4. Proof of Theorem 6.5. Fix an episode boundary and an intense point x , classified at its active scale by Section 14.1.

Case 1: organizing. The active ball is coherent with some gap g . If g is nondegenerate, Section 14.2 aligns the point within the fast time $T_c = g^{-1} \log(1/\theta^*)$ and Section 14.3 then collapses the aligned eigenvalue at intensity: the point exits to the capped horn carrying tube structure (C1)–(C2). If the gap degenerates, the pancake fork of Section 14.3 excludes persistence. Either way, an organizing point that sustains production becomes a class tube (or is capped) within the fast clock.

Case 2: capped. By Definition 6.2 the trailing-window average of the pairing is at most $\lambda_2 + m$; at the cap’s margin the production is $\leq (\lambda_2 + m)$ -paced, and λ_2 at tube-dominated points is the vetoed axial eigenvalue (zero at leading order, `ft_middle_eigenvalue`) while at background-dominated points it is the budgeted quantity of Lemma 7.2: capped points are productionless up to the slow clock and the counted margin.

Case 3: arranged-tangled. By Section 14.1(iv) the point satisfies the multi-window maintenance hypothesis — exactly the funnel’s input, excluded in Section 9 (Theorem 9.4).

Measurability and bookkeeping. Each horn is defined by measurable conditions (time-averages of continuous quantities; sups over dyadic scales), so the horns partition the intense set measurably; the Besicovitch subfamily of Section 14.1(iii) carries the counts; and every exit, culling, and merger event is logged by the episode ledger of Section 7, whose costs are priced there. Consequently at every episode boundary each intense structure sustaining production is class-organized or capped-productionless, and the hypotheses of Theorem 5.1 are re-supplied. $\square$

15. COMPLETE PROOFS FOR SECTION 7: THE STRAIN BUDGET

15.1. Proof of Lemma 7.1. (i) *The vetoed pair kernel.* Let two distinct class tubes have separation $d \geq R_m \max(\delta_i, \delta_j)$ (clause (C3): closer pairs are one structure), axis angle θ_{ij} , and let κ_j bound the companion’s curvature. Near the closest-approach segment, model the companion by its local filament and write its transverse displacement from the straight parallel line as $h(z)$, $|h(z)| \leq \frac{1}{2}\kappa_j z^2$ on $|z| \lesssim d$. To first order in h the companion’s velocity field is the *shifted two-dimensional field*

$$u(x_\perp, z) = u_{2D}(x_\perp - h(z)),$$

where u_{2D} is the straight tube’s planar field: each z -slice is the same solenoidal planar field, merely translated. This ansatz has zero divergence, and its curl is the tilted core to $O((\kappa_j d)^2)$ — but Biot–Savart is determined by the curl, so the companion’s actual

field differs from the ansatz only at $O((\kappa_j d)^2) \cdot \Gamma_j/d$. The ansatz has $u_z \equiv 0$, hence its axial strain component vanishes *identically*: $S_{33} = \partial_z u_z \equiv 0$ (machine-certified in matrix form, `cp_veto_shifted_field`). Its off-axis entries S_{13}, S_{23} are first order in the tilt h' , of size $O(\kappa_j d \cdot \Gamma_j/d^2)$; and in the pairing of tube i , whose axis makes angle θ_{ij} with the companion's, the expansion of $\xi^\top S \xi$ carries *no* $\cos^2 \theta$ term when $S_{33} = 0$ (machine-certified, `cp_axial_residue`): every surviving term is at least first order in $\sin \theta_{ij}$ against entries themselves first order in tilt, or second order outright. Collecting orders,

$$\alpha_{ij} \leq C_v (\theta_{ij} + \kappa_j d)^2 \frac{\Gamma_j}{d^2},$$

for parallel and anti-parallel geometry alike (θ is taken mod π), with C_v from the kernel constants of Section 13.2. $\square$ (i)

(ii) *The equilibrium payment.* Call the pair *mutually dominated* when $\Gamma_j/d^2 \geq \gamma$, i.e. $d \leq d_{\gamma,j} = (\Gamma_j/\gamma)^{1/2}$. There the mutual strain is the organizing strain, with gap $g_{\text{pair}} = c_g \Gamma_j/d^2$, and Section 14.2 pins a persistent pair at

$$\theta_{\text{eq}} \leq C \frac{\gamma}{\Gamma_j/d^2} = C \frac{\gamma d^2}{\Gamma_j}$$

— the equilibrium tightens with the pair's own intensity. By (i) at equilibrium, the angular part of the payment is

$$C_v \theta_{\text{eq}}^2 \frac{\Gamma_j}{d^2} = C_v C^2 \frac{\gamma^2 d^2}{\Gamma_j} \leq C_v C^2 \gamma \quad (d \leq d_{\gamma,j}),$$

(the evaluation at the dominance edge is machine-certified, `cp_payment_algebra`). Summing over the band with the enstrophy-weighted count: each class tube carries $E_j = c_E \Gamma_j^2 L_j / \delta_j^2 \geq c_E \Gamma_{\min}^2 L_{\min} / \delta_c^2$, where by the strain band of (C2) the core ceiling is $\delta_c^2 \leq 4\nu/(c_1 \gamma)$ (a tube below the band fattens past δ_c , falls below ω^* , and is culled — clause (C5)); hence

$$N_{\text{band}} \leq \frac{\Omega \delta_c^2}{c_E \Gamma_{\min}^2 L_{\min}} \leq \frac{4\nu \Omega}{c_1 \gamma c_E \Gamma_{\min}^2 L_{\min}}, \quad \sum_{\text{band}} d_{ij}^2 \leq N_{\text{band}} \cdot \frac{\Gamma_{\max}}{\gamma},$$

and the band total is

$$\sum_{\text{band}} \alpha_{ij} \leq C_v C^2 \frac{\gamma^2}{\Gamma_{\min}} \cdot N_{\text{band}} \cdot \frac{\Gamma_{\max}}{\gamma} = C_n \nu \Omega(t), \quad C_n = \frac{4C_v C^2 \Gamma_{\max}}{c_1 c_E \Gamma_{\min}^3 L_{\min}} :$$

the γ 's cancel identically (machine-certified, `cp_band_cancellation`). The bend part of (i) contributes $C_v (\kappa_j d)^2 \Gamma_j/d^2 = C_v \kappa_j^2 \Gamma_j \leq c \omega_j/M^2$ by (C1) — a small-class-constant multiple of the core scale, paid into the collapse margin of Section 14.3 exactly as its θ^{*2} term, with the same one-directional choice of M . Pairs in the ambient-dominated band $d \in [d_{\gamma,j}, \ell_N]$ contribute at most γ each by the definition of $d_{\gamma,j}$ and are counted by the same enstrophy weight, landing in the same $C_n \nu \Omega(t)$ form. $\square$ (ii)

(iii) *Transients.* A pair not at equilibrium (a crossing) contracts toward it at the mutual organizer's rate $g_{\text{pair}} = c_g \Gamma_j/d^2$ by Section 14.2, while its unvetoes payment is at most $\theta^2 \Gamma_j/d^2$: the integrated cost of one transient is

$$\int \theta(t)^2 \frac{\Gamma_j}{d^2} dt \leq \theta_0^2 \frac{\Gamma_j}{d^2} \cdot \frac{1}{2g_{\text{pair}}} = \frac{\theta_0^2}{2c_g} \leq C_t,$$

an absolute constant — the organizer's own magnitude sets the contraction clock and cancels. Events are finitely many per tube per episode: concurrent band partners number

at most the Besicovitch constant b_3 ; re-entry into the band is driven by the ambient strain (slow clock, rate γ , against episode length c/γ : $O(1)$ re-entries); and a merger ends both parents' identities (clause (C3)), so a given identity merges at most once. Total per-episode transient cost $\leq C_{\text{pair}} = b_3 C_{\text{re}} C_t + C_{\text{merge}}$, absorbed multiplicatively in the episode chain: $C'_1 \rightarrow C'_1 + C_{\text{pair}}/c$. $\square$ (iii)

(iv) *The far field, over cells.* Partition $\mathbb{T}^3$ into cells of side $\ell_N = c_N N^{-1/3}$ — the mean spacing, correct by pigeonhole from the count $N \leq \Omega/E_{\min}$; *cells pack by construction, and no minimum-separation hypothesis on tubes enters anywhere.* For a cell Q at dyadic shell distance $\sim 2^k \ell_N$ from tube i , the kernel bound of Section 13.2 gives $|S[\omega \mathbf{1}_Q](x_i)| \leq C \ell_N^{3/2} (2^k \ell_N)^{-4} \|\mathbf{u}\|_{L^2(Q)}$; the shell holds at most $c \cdot 2^{3k}$ cells; Cauchy-Schwarz over the cells of each shell and geometric summation give

$$\gamma_i^{\text{far}} \leq C_\gamma \left(\sum_{k \geq 0} 2^{3k} \cdot \ell_N^3 (2^k \ell_N)^{-8} \right)^{1/2} \|\mathbf{u}_0\|_{L^2} = C_\gamma \left(\frac{4}{3} \right)^{1/2} \ell_N^{-5/2} \|\mathbf{u}_0\|_{L^2},$$

the series $\sum 4^{-k} = \frac{4}{3}$ machine-certified (`w9_shell_series`) and $\|\mathbf{u}\|_{L^2} \leq \|\mathbf{u}_0\|_{L^2}$ by the energy identity. Sub- ℓ_N vorticity is the near band, already paid by (ii)–(iii) and by Section 15.2. $\square$

15.2. Proof of Lemma 7.2. (i) *The threshold and the log bound.* The threshold $\omega^* = c_* \omega_{\text{rms}}(t)$ is a measurable function of the solution's present state and never depends on γ : the feared self-reference is severed at the definition. Decompose $\omega = \omega_{\text{class}} + \omega_{\text{cx}} + \omega_b$ with $|\omega_b| \leq \omega^*$; every unit of vorticity lies in exactly one part. The budget consumes *core-averaged* strain, and averaging against the core profile mollifies the Biot-Savart strain kernel at scale δ_i : $|K_{\delta_i}(z)| \leq C \min(\delta_i^{-3}, |z|^{-3})$, with the spherical cancellation retained for $|z| \geq \delta_i$. Split B_{ℓ_N} into dyadic shells from δ_i to ℓ_N ; on each shell Cauchy-Schwarz gives the switch

$$|\langle S[\omega_b \mathbf{1}_{\text{shell}}] \rangle_i| \leq C (2^{-j} \ell_N)^{-3/2} \min(\omega^* \text{vol}^{1/2}, (2\Omega)^{1/2}),$$

and the L^∞ branch pays exactly $C\omega^*$ per shell: summing the $\log_2(\ell_N/\delta_i)$ shells,

$$|\langle S[\omega_b \mathbf{1}_{B_{\ell_N}}] \rangle_i| \leq C_b \omega^* \log(e + \ell_N/\delta_i);$$

the sub- δ_i contribution is one mollified scale ($\leq C\omega^*$, no logarithm), and beyond ℓ_N the background rides inside the cells of Section 15.1(iv). $\square$ (i)

(ii) *The absorption, with structural coefficient.* By the Burgers band, $\delta_i \geq (4\nu/C'_1 \gamma)^{1/2}$, so γ enters the bound only inside the logarithm: the budget inequality has the shape $\gamma \leq A + B \log(e + C_\delta \gamma)$ with A, B as in Theorem 7.3. The elementary factorization

$$e + C_\delta \gamma \leq (e + C_\delta(A + B)) \left(1 + \frac{\gamma}{A + B} \right)$$

holds identically (the surplus is $C_\delta(A + B) + e\gamma/(A + B) > 0$), and with $\log(1 + t) \leq t$ the logarithm splits so that the coefficient of γ on the right is $B/(A + B) < 1$ — *structural, not tuned*. The absorbed form

$$A\gamma \leq (A + B)(A + B \log(e + C_\delta(A + B)))$$

is machine-certified (`bg_log_absorption`). $\square$ (ii)

(iii) *The intense sub-circulation residue.* Structures with $|\omega| > \omega^*$ but $\Gamma < \Gamma_{\min}$ are of two kinds. *Culled remnants* (exits and reconnection debris) are born in events already counted by Section 15.1(iii); they are productionless (Theorem 6.5: no sustained pairing without organization), so they decay on their viscous core time δ^2/ν , and their integrated strain

on tube i over that lifetime is at most $C\Gamma_{\min}/(\nu R_m^2)$ per event — an absolute constant, absorbed into C_{pair} . *Wrapped satellites* — vorticity wound azimuthally about the tube, the coaxial-ring geometry whose axial push the parallel veto does not suppress (a ring drives axial flow on its axis; cf. the core-dynamics phenomenology [MH94]) — accumulate at most a ledger-bounded circulation $\Gamma_{\text{sh}} \leq C_{\text{pair}}\Gamma_{\min}$ per episode (each capture is a counted event, and the sheath turns over viscously within the episode); their axial strain is then

$$\frac{\Gamma_{\text{sh}}}{(R_m\delta_i)^2} \leq \frac{C_{\text{pair}}\Gamma_{\min}C'_1}{4\nu R_m^2} \gamma = \lambda\gamma,$$

using the Burgers relation $\delta_i^{-2} \leq C'_1\gamma/4\nu$ — a *multiplicative* term. The merger-floor constant R_m is free, data-only, and chosen with $\lambda \leq \frac{1}{2}$; the absorption $\gamma \leq X + \lambda\gamma \Rightarrow \gamma \leq 2X$ is machine-certified (`bg_half_absorption`). $\square$

15.3. Proof of Theorem 7.3. *The pointwise display.* Collecting: the far field (Section 15.1(iv)) contributes $K'\Omega^{5/6}$ after inserting $\ell_N \geq c_N(E_{\min}/\Omega)^{1/3}$ (the exponent chain $(x^{-1/3})^{-5/2} = x^{5/6}$ is machine-certified, `w10_exponent_chain`); the near band (Section 15.1(ii)–(iii)) contributes $C_n\nu\Omega$; the background log bound (Section 15.2(i)) contributes $B\log(e + C_\delta\gamma)$ with $B = C_b\omega^*$; and the sheath contributes $\lambda\gamma$, $\lambda \leq \frac{1}{2}$. Absorbing the sheath first ($\gamma \leq 2X$ form) and then the logarithm (Section 15.2(ii)) yields

$$\gamma(t) \leq 2A(t) + 2B(t)[1 + \log(e + C_\delta(A(t) + B(t)))], \quad A = K'\Omega^{5/6} + C_n\nu\Omega.$$

Time integration. Hölder in time with the conjugate pair $(\frac{6}{5}, 6)$ — the pairing $\frac{5}{6} + \frac{1}{6} = 1$ is machine-certified (`w10_holder_exponents`) — against the one unconditional integral $\int_0^\infty \Omega dt \leq E_0/2\nu$ gives $\int_0^T K'\Omega^{5/6} dt \leq KT^{1/6}$; the near band integrates directly, $\int_0^T C_n\nu\Omega dt \leq C_nE_0/2 =: K_2$; and the background term obeys the pointwise bound $\omega^*\log(e + \cdot) \leq 2\Omega^{5/6} + 3$ in class units (its $\Omega^{5/6}$ part joins K , its constant part integrates to K_3T). Hence

$$\int_0^T \gamma dt \leq KT^{1/6} + K_2 + K_3T \quad \text{for every } T < \infty,$$

with K, K_2, K_3 explicit in (E_0, ν) and class constants.

The drift lemma. Within an episode the budget's sources drift by bounded factors: production is *pairing-capped* — $d\Omega/dt = P - D$ with P concentrated on class cores where $\alpha \leq C'_1\gamma$ (Section 13.11), null on the background (Theorem 6.5), and event-bounded on transients (Section 15.1(iii)) — so $d\Omega/dt \leq 2C'_1\gamma\Omega + \text{event terms}$ and over the episode length c/γ every budgeted quantity moves by at most $e^{C_d c}$; hence $\gamma(t) \leq e_d\gamma(t_n)$ throughout, absorbed multiplicatively into the episode constant. $\square$

16. COMPLETE PROOFS FOR SECTION 8: THE CHAIN AND THE CRITERION

16.1. Proof of Lemma 8.1. *The forced-damped balance.* The elliptical deformation of a class tube's core is the $m = 2$ mode of the local linear theory of Section 12 about the modulated column. Two rates compete. The ambient strain forces the mode at $\|F_{m=2}\| \leq C_f\gamma$ (the forcing decomposition of Section 13.11, Step 2, restricted to the $m = 2$ sector). The damping is the column's own: purely inviscid mixing of an $m = 2$ disturbance is algebraic [BCV19; Sch+00], but at the Burgers core the *viscous critical layer* damps the quasi-mode exponentially at the classical rate $\sim \omega_0 \text{Re}_\Gamma^{-1/3}$, where $\text{Re}_\Gamma = \Gamma/\nu$ is the circulation Reynolds number. The structural fact that saves the fast clock: for class tubes,

(C2) bounds $\text{Re}_\Gamma \in [\Gamma_{\min}/\nu, \Gamma_{\max}/\nu]$ — a *class parameter*, not a growing quantity — so the damping rate has the floor

$$c_2 \geq c(\nu/\Gamma_{\max})^{1/3}$$

(the antitone bound $x \leq B \Rightarrow x^{-1/3} \geq B^{-1/3}$ is machine-certified, `ec_rate_floor`). The ellipticity therefore obeys

$$\frac{de}{dt} \leq -c_2 \omega_0 e + C_f \gamma, \quad e(t) \leq e(0) e^{-c_2 \omega_0 t} + e_{\text{eq}}, \quad e_{\text{eq}} = \frac{C_e \gamma}{\omega_0}, \quad C_e = \frac{C_f}{c_2},$$

by the explicit solution of the linear comparison equation — the equilibrium ellipticity vanishes linearly in the clock ratio γ/ω_0 , in agreement with the classical strained-vortex equilibria at weak strain [MS71; Kid81].

The perturbative bootstrap. The $m = 2$ linear theory is valid for $e \leq 1/M$. For tubes in the load-bearing regime $\omega_0 \geq R_7 \gamma$ with $R_7 = 2MC_e$,

$$e_{\text{eq}} = \frac{C_e \gamma}{\omega_0} \leq \frac{C_e \gamma}{2MC_e \gamma} = \frac{1}{2M}$$

(machine-certified, `ec_threshold`) — strictly inside the validity window — and since the relaxation is a contraction at the fast rate $c_2 \omega_0$, a tube entering the regime with $e \leq 1/M$ is pulled to $e \leq 1/(2M)$ within the fast time and held there: the bootstrap closes, and clause (C1)'s near-circularity in the load-bearing regime is *derived*. Below the clock ratio no statement is made — none is needed, by the split of Section 16.3. $\square$

16.2. Proof of Lemma 8.2. Fix a flow trajectory $X(t)$ and write $\omega(t) = |\omega(X(t), t)|$, so $\frac{d}{dt}\omega = \alpha(X(t), t)\omega$. By Section 14.1 the trajectory's time partitions measurably into intervals of three kinds, classified at the active scale.

Class residence. While X rides a class tube, $\alpha \leq C'_1 \gamma$ by Section 13.11 — the cap is pointwise on the core.

Capped intervals. By Definition 6.2 the trailing-window pairing average is at most $\lambda_2 + m$. The middle eigenvalue's size splits by regime: at tube-dominated points the local strain is two-dimensional to leading order, with eigenvalues $(\sigma, 0, -\sigma)$ — the middle one is the vetoed axial eigenvalue, zero at leading order (machine-certified, `ft_middle_eigenvalue`) — so $\lambda_2 \leq c_\lambda \gamma$ plus the vetoed corrections; at background-dominated points $\lambda_2 \leq \|S_{\text{bg}}\|$, which is exactly the quantity budgeted in Section 15.2. Either way the interval's growth factor is $e^{(\lambda_2 + m)\Delta t}$ with $\int \lambda_2$ paid by the repaired budget.

Funnel transits. An occupancy of the organizing or tangled horns that ends in class entry or capping is a transit. By Section 14.2 the organizing strain of gap g contracts the misalignment within $T_c = g^{-1} \log(1/\theta^*)$, while the growth rate during the transit is at most λ_1 . If the gap is nondegenerate, $\lambda_1 \leq g/c_o$ (the organizer bounds its own top eigenvalue on coherent patches); the per-transit growth factor is then

$$\exp(\lambda_1 T_c) \leq \exp\left(\frac{g}{c_o} \cdot \frac{\log(1/\theta^*)}{g}\right) = \theta^{*-1/c_o}$$

— *independent of the intensity and of g* : the organizing strain sets both the growth and its own contraction clock, and g cancels (machine-certified, `ft_transit_factor`). Gap-degenerate transits are pancake-type and are excluded by the fork of Section 14.3. Occupancies that terminate in neither class entry nor capping would persist in the tangled horn; persistent tangled occupancy at any point is impossible by Theorem 9.4, so every occupancy is a transit.

Counting and the fold. After a transit the point is class or capped; re-entry into the funnel requires re-tangling, which is driven by the ambient differential strain at the slow rate $c_r\gamma$; against episode length c/γ this gives at most $c_r \int \gamma dt + 1$ transits per episode. Multiplying the factors along the partition and folding ($e^A G^n = e^{A+n \log G}$, machine-certified, `ft_chain_fold`),

$$\omega(T) \leq \omega(0) \exp \left[(C'_1 + c_\lambda) \int_0^T \gamma dt + mT + \frac{1}{c_o} \log(1/\theta^*) \left(c_r \int_0^T \gamma dt + N_{\text{ep}} \right) \right],$$

every γ -integral paid by Section 15.3 and N_{ep} finite on finite intervals. $\square$

16.3. Proof of Theorem 8.3. Let every episode boundary in $[a, b)$ present all intense structures class-organized or capped. The pointwise identity

$$\max(w - r, 0) + \min(w, r) = w$$

(machine-certified, `ec_bkm_split`) splits the vorticity supremum at the clock-ratio threshold $r = R_7\gamma(t)$:

$$\int_a^b \|\omega\|_\infty dt = \int_a^b \min(\|\omega\|_\infty, R_7\gamma) dt + \int_a^b (\|\omega\|_\infty - R_7\gamma)_+ dt.$$

The first integral is at most $R_7 \int_a^b \gamma dt \leq R_7[K(b-a)^{1/6} + K_2 + K_3(b-a)]$ by Section 15.3, *with no appeal to the linear theory*: below the clock ratio, cores may be as elliptical as they like. On the second integral, any trajectory attaining $\|\omega\|_\infty > R_7\gamma$ is intense-class at the attaining point, provably near-circular by Section 16.1, and rides the chain of Section 16.2 with base value at most $\max(\|\omega(a)\|_\infty, R_7\gamma(a))$; the organization hypothesis re-supplies Section 13.11 at every boundary in $[a, b)$, and the within-episode drift is absorbed by the drift lemma of Section 15.3. Chaining episodes yields the display of Theorem 8.3, finite for $b < \infty$; hence $\int_a^b \|\omega\|_\infty dt < \infty$, and the Beale–Kato–Majda criterion [BKM84] extends smoothness past b . $\square$

17. COMPLETE PROOFS FOR SECTION 9: THE FUNNEL AND THE SQUEEZE

17.1. Proof of Lemma 9.1. (i) *The Morrey identity and the data bound.* With $\Phi_r(t) = r \mathbf{u}(x^* + ry, t)|_{B_1}$, the substitution $x = x^* + ry$ gives exactly

$$\int_{B_1} |\Phi_r|^2 dy = r^2 \cdot r^{-3} \int_{B_r(x^*)} |\mathbf{u}|^2 dx = r^{-1} \int_{B_r(x^*)} |\mathbf{u}|^2 dx$$

(the exponent arithmetic $r^2 \cdot r^{-3} = r^{-1}$ is machine-certified, `sc8_slice_scaling`): the critical Morrey density at scale r is the squared $L^2(B_1)$ norm of the amplitude-carrying slice. The density is continuous on L^3 -bounded sets of slices, $\|\Phi\|_{L^2(B_1)} \leq |B_1|^{1/6} \|\Phi\|_{L^3(B_1)}$ (Hölder with $\frac{1}{2} = \frac{1}{3} + \frac{1}{6}$, machine-certified, `sc8_holder_exponents`), so the maintenance recurrence transports the first window's value: $\mathfrak{M}(r_k)^{1/2} \leq \mathfrak{M}(r_0)^{1/2} + |B_1|^{1/6} C\varepsilon$, whence $\sup_k \mathfrak{M}(r_k) \leq e^{2C} \mathfrak{M}(r_0)$. Between windows the per-window gain e^C interpolates with the dyadic doubling factor λ_0^{-1} :

$$\mathfrak{M}_0 = \sup_{0 < \rho \leq r_0} \mathfrak{M}(\rho) \leq \lambda_0^{-1} e^{4C} \mathfrak{M}(r_0) < \infty,$$

and at the outer window the global energy identity gives the data-only bound $\mathfrak{M}(r_0) \leq 2E_0 + \nu^{-1}E_0$ at $r_0 = 1$.

(ii) *Compactness, suitability, and the floor.* With $\mathfrak{M}_0 < \infty$ the rescalings $u^{\lambda_k}(y, s) = \lambda_k \mathbf{u}(x^* + \lambda_k y, T^* + \lambda_k^2 s)$, $\lambda_k = r_k \rightarrow 0$, are uniformly bounded in $L_t^\infty L_x^2 \cap L_t^2 H_x^1$ locally by scaling invariance; the pressure is controlled by the local decomposition $p = p_{\text{loc}} + p_{\text{harm}}$ of local Leray theory [Lem02; KS07]; Aubin–Lions gives strong L_{loc}^3 convergence of a subsequence; and suitability passes to strong L^3 limits [Lin98; Sch76; CKN82]. The limit V is an ancient suitable solution of the self-similar system. For the floor: at a singular point every parabolic cylinder carries $\iint_{Q_\rho} (|\mathbf{u}|^3 + |p|^{3/2}) \geq \varepsilon_{\text{CKN}} \rho^2$ [CKN82; Lin98]; in similarity variables each unit time-slab of the orbit is a rescaled cylinder at the singular point, so the bound holds uniformly along the orbit, and strong convergence transports it to every ω -limit member: $\iint_{Q_1} |V|^3 \geq \varepsilon_{\text{CKN}}$ member-wise.

(iii) *The limit set.* The orbit $\{v(\tau)\}$ is precompact in X_{loc} by (ii) at every τ , so $\mathcal{W} = \bigcap_\sigma \{v(\tau) : \tau \geq \sigma\}$ is nonempty and invariant under the (Leray-system) flow, and every member inherits the recurrence bound: $\text{dist}(V(\tau), V(\tau + T)) \leq C\varepsilon$ for all τ , with $T = 2 \log(1/\lambda_0)$. $\square$

17.2. Proof of Lemma 9.2. (i) *Confinement by two pins.* Let Φ_0 be the recurrent profile and $\delta(\tau) = V(\tau) - \Phi_0$ on a window $[\tau_0, \tau_0 + T]$; the maintenance recurrence pins $\|\delta(\tau_0)\|, \|\delta(\tau_0 + T)\| \leq C\varepsilon$. Linearizing the self-similar flow along Φ_0 , $\delta' = \mathcal{L}\delta + N(\delta, \delta)$ with $\|N(\delta, \delta)\| \leq C_N \|\delta\|^2$ (the bilinear estimate of Section 13.7 in the class frame). Because Φ_0 sustains the window-averaged pairing (the funnel’s content), the deviation space splits into bundles carried by proved statements: the symmetry modes are removed by the modulation quotient (Section 13.5: the Gram is nondegenerate); the aligning cone contracts at the gap of Section 14.2; the production cone expands at the exact 3:1 rate; and the exterior multipoles are slaved by Section 13.10. No other neutral directions exist at a pairing-sustaining state — that is the content of the gap $g > 0$ in Section 14.2. The two-pin shadowing argument then confines the orbit: contracting components are controlled *forward* from the left pin, $\|\delta_s(\tau)\| \leq e^{-g(\tau-\tau_0)} C\varepsilon + (\text{nonlinear})$; the expanding component is controlled *backward* from the right pin — expansion forward is contraction backward, $\|\delta_u(\tau)\| \leq e^{-3\alpha_{\min}(\tau_0+T-\tau)} C\varepsilon + (\text{nonlinear})$ — *a deviation that is small at the window’s end and expands in between was never large; nothing restores anything*. The window length $T = 2 \log(1/\lambda_0)$ is fixed and every rate is bounded by the class constant $\alpha_{\max}(M_0)$ in similarity variables, so within-window excursions cost at most the fixed factor $e^{3\alpha_{\max}T}$, absorbed into the constant; the nonlinear terms close by the bootstrap for $\varepsilon \leq \varepsilon_0$ ($C_N(2C_s\varepsilon)^2 < C_s\varepsilon$). Chaining windows — each hands the next its pin — gives $\text{dist}(V(\tau), \Phi_0) \leq C_s\varepsilon$ for all τ .

(ii) *The derived defect.* The vector field $F(v) = \Delta v - \frac{1}{2}v - \frac{1}{2}(y \cdot \nabla)v - \mathbb{P} \text{div}(v \otimes v)$ is Lipschitz from L_{loc}^3 to H_{loc}^{-2} on class-bounded sets: the linear terms are bounded $L_{\text{loc}}^2 \rightarrow H_{\text{loc}}^{-2}$ (the y -weight harmless on bounded sets); the quadratic term satisfies $\|\text{div}(v \otimes v - w \otimes w)\|_{H^{-2}} \leq C\|(v-w) \otimes (v+w)\|_{L^{3/2}} \leq C\|v-w\|_{L^3}(\|v\| + \|w\|)_{L^3}$ with $L_{\text{loc}}^{3/2} \hookrightarrow H_{\text{loc}}^{-1}$ in three dimensions; and the pressure gradient is controlled through the local decomposition of Section 17.1(ii), the harmonic part’s interior variation bounded along the confined orbit by mean-value estimates. Time-averaging one window: $\frac{1}{T} \int_{\tau_0}^{\tau_0+T} \partial_\tau V d\tau = (V(\tau_0 + T) - V(\tau_0))/T$ has size $\leq 2C_s\varepsilon/T$ by confinement, while $\frac{1}{T} \int_{\tau_0}^{\tau_0+T} F(V) d\tau = F(\Phi_0) + O(\varepsilon)$ by the Lipschitz property; equating, $\|F(\Phi_0)\|_{H_{\text{loc}}^{-2}} \leq C\varepsilon$ — the fixed profile is ε -approximately stationary — and pointwise, for every τ , $\|\partial_\tau V(\tau)\|_{H_{\text{loc}}^{-2}} = \|F(V(\tau))\| \leq \|F(\Phi_0)\| + C\varepsilon \leq C_1\varepsilon$. On the class’s smooth regions (the far-field expansion region and the ε -regular interior, with class-uniform C^k bounds by ε -regularity and Schauder), interpolation lifts the weak defect at square-root

cost, $\|\partial_\tau V\|_{L^2(K)} \leq C\|\partial_\tau V\|_{H^{-2}}^{1/2}\|\partial_\tau V\|_{H^2}^{1/2} \leq C(M_0)\varepsilon^{1/2}$, and the class's linear-in- R energy growth matches the $R^{1/2}$ normalization on large balls: the defect in the quantitative-rigidity norm is $\varepsilon' = C_2\varepsilon^{1/2}$. $\square$

17.3. Proof of Lemma 9.3. Tsai's scheme [Tsa98; NRŠ96], run with the defect carried through each step.

The localized energy identity. Testing the self-similar system with $V\varphi_R^2$, the scaling terms combine exactly: $\frac{1}{2}\int|V|^2\varphi^2 + \frac{1}{2}\int(y \cdot \nabla V) \cdot V\varphi^2 = -\frac{1}{4}\int|V|^2\varphi^2 + \text{tails}$, giving

$$\int|\nabla V|^2\varphi_R^2 - \frac{1}{4}\int|V|^2\varphi_R^2 = -\int(\partial_\tau V) \cdot V\varphi_R^2 + T_R,$$

$$\left|\int(\partial_\tau V) \cdot V\varphi^2\right| \leq C\varepsilon' R^{1/2}\|V\|_{L^2(B_{2R})}^{\text{norm}},$$

the exact $-\frac{1}{4}$ balance machine-certified (`qt1_scaling_coefficient`, `qt1_energy_identity`), and T_R the annulus tails — viscous, cubic, and pressure terms on $\{R \leq |y| \leq 2R\}$, each bounded by the suitable-class annulus norms via Gagliardo–Nirenberg interpolation (bounded raw; refined below).

The expansion along rays. In the log-radial variable along a ray, the profile equation is a constant-characteristic linear system perturbed by the nonlinearity and the defect, of integrable relative size $O(e^{-u})$; Levinson's asymptotic integration theorem [CL55] applies, giving $V = a_{\varepsilon'}(\hat{y})/|y| + O(|y|^{-3/2})$ with coefficients continuous in the data and constants uniform in $\hat{y}$ by compactness of the sphere. With the expansion, the annulus tails refine to $T_R \leq C(|a_{\varepsilon'}|^2 + o_R(1))$, vanishing along $R \rightarrow \infty$ at the order of the asymptote.

The head pressure. $\Pi = |V|^2/2 + q$ satisfies the Nečas–Růžička–Šverák elliptic structure with an $O(\varepsilon'|V|)$ source; the maximum principle degrades quantitatively, $\sup \Pi \leq \limsup_{|y| \rightarrow \infty} \Pi + C\varepsilon'$, and the annulus-averaged decay of Π at infinity follows from the expansion and the far-field pressure representation.

The flux balance and the honest exponent. At $\varepsilon' = 0$ the boundary flux of the energy identity forces $a_0 \equiv 0$ (this is the rigidity of [NRŠ96; Tsa98]); the flux is *quadratic* in the asymptote, so the defect version gives $|a_{\varepsilon'}|^2 \leq C\varepsilon'$, i.e. $|a_{\varepsilon'}| \leq C\varepsilon'^{1/2}$ — the exponent is $\frac{1}{2}$, not 1. Feeding the expansion, the Π -principle, and the energy identity back into one another, the balance closes linearly in the remaining data: $\|V\|_{E(B_2)} \leq C\varepsilon'^{1/2}$. Discretely self-similar variants [Tsa14; CW17] cover any weakened reading of the maintenance windows. $\square$

17.4. Proof of Theorem 9.4. Chain the exponents: the maintenance tolerance ε gives the confinement and the derived defect $\varepsilon' = C_2\varepsilon^{1/2}$ (Section 17.2); the quantitative rigidity gives $\|V\|_{E(B_2)} \leq C\varepsilon'^{1/2} = C'\varepsilon^{1/4}$ (Section 17.3); and the parabolic interpolation $\|V\|_{L^{10/3}(Q_1)} \leq C\|V\|_{L^\infty L^2}^{2/5}\|V\|_{L^2 H^1}^{3/5} \leq C\|V\|_{E(B_2)}$ gives

$$\iint_{Q_1} |V|^3 \leq C_{QT} \varepsilon^{3/4}.$$

Choose the class tolerance $\varepsilon < \varepsilon_7 = (\varepsilon_{\text{CKN}}/C_{QT})^{4/3}$; then $C_{QT}\varepsilon^{3/4} < \varepsilon_{\text{CKN}}$ (machine-certified, `sc7_small_eps_choice`), contradicting the member-wise floor of Section 17.1(ii). No member of $\mathcal{W}$ exists; since a nonempty $\mathcal{W}$ is guaranteed whenever the maintenance hypothesis holds (Section 17.1(iii)), the hypothesis itself is unsatisfiable: persistent arranged-tangled occupancy at a singular point is impossible.

The order of quantifiers, displayed once more. ε_{CKN} is universal [CKN82; Lin98]; (E_0, ν) are the data; $\mathfrak{M}(r_0)$ and M_0 are data-only (Section 17.1(i)); $C_{\text{QT}} = C_{\text{QT}}(M_0)$ is data-only; ε_7 is data-only; and the pairing margin m is chosen small enough that $\varepsilon(m) < \min(\varepsilon_0, \varepsilon_7)$ — a choice made, like every entry of Table 2, before any solution trajectory or singular time is examined. No lower bound on ε exists anywhere in the argument. $\square$

APPENDIX A. THE MASTER CONSTANTS TABLE

Table 2 extends Table 1 with the constraints. Reading: each tier depends only on tiers above it and the data (E_0, ν) ; the graph is acyclic; every “large”/“small” constraint is one-directional; solution-dependent quantities appear only in estimates, never in the choice of a constant.

tier	constants	constraints and provenance
0	$\varepsilon_{\text{CKN}}, b_3, \text{Riesz}$	universal: the CKN ε -regularity constant [CKN82; Lin98]; the Besicovitch overlap [Mat95]; the Hilbert-transform bound
1	E_0, ν	the data
2	$\mathfrak{M}(r_0) \leq 2E_0 + \nu^{-1}E_0$ $M_0 = \lambda_0^{-1}e^{4C}\mathfrak{M}(r_0)$; $\alpha_{\max}(M_0)$ $\Gamma_{\max}(E_0)$	the energy identity; the window interpolation of Lemma 9.1(i)
3	$\Delta_0 \leq \frac{1}{3}$; $\lambda_0 = \frac{1}{2}$; $T = 2\log(1/\lambda_0)$ $c_*, c_1, c_E, c_\lambda, c_o, c_r, C_f$	fixed by the linear theory and the definitions; $\Delta_0 \leq \frac{1}{3}$ is the inheritance condition of Lemma 6.1
4	$c_2 \geq c(\nu/\Gamma_{\max})^{1/3}$; $C_e = C_f/c_2$ $C_{\text{QT}}(M_0)$; C_{su}	derived: the bounded circulation Reynolds number (Lemma 8.1); quantitative rigidity (Lemma 9.3); sector uniformity (Lemma 4.4)
5	M large; θ^* small	free, one-directional: the perturbative windows of Sections 4 and 5
6	$R_7 = 2MC_e$; ω_7	large: the circularization bootstrap (Lemma 8.1); subsumes the collapse and transient-dominance thresholds
7	R_m large ($\lambda \leq \frac{1}{2}$ in Lemma 7.2); R_v	large: the merger floor; the sheath absorption
8	m small: $\varepsilon(m) < \min(\varepsilon_0, \varepsilon_7)$, $\varepsilon_7 = (\varepsilon_{\text{CKN}}/C_{\text{QT}})^{4/3}$	small: the squeeze (Theorem 9.4); no lower bound on ε exists anywhere in the argument

TABLE 2. The dependency order of the named constants, with constraints.

APPENDIX B. THE CLAIM MATRIX

The development was tracked against a claim matrix: one row per component, each row closed only when its statement reached referee-grade prose with its exact core certified. The matrix below is the closed state; L = Lean certificate (Appendix C), S = symbolic adjudication log, I = calibrated instrument.

component	status	certificates
financing identity (with sign)	exact	L×4, S, I
pancake discriminant identity	exact	L, I

component	status	certificates
veto; parity kernel	exact	L×2, I
Rayleigh-monotone stability	full	L (core)
2D band: monotone reduction, annulus, depletion, edge	full	I series
viscous band (matched layer, parametrix)	full	I (rate 0.387)
$m = 1$ sector (translation pair, impulse adjoint)	full	L (Gram), I
sector uniformity; quantified displays	full	L×3, S
organization–production (pairing; null background)	full/exact	L×3, I
alignment rigidity; veto–parity collapse	full	I (basin)
budget identity; moment law; calibration constant	exact	L×5, S
episode estimates (kernel, parity, gain, modulation, adiabatic, bilinear)	full	L×8, S
episode cap $\alpha \leq C'_1\gamma$; smoothing; local existence	full	L ($\beta = \pi$), S
covering and counting; the cell far field	full	L (series), S
close pairs (vetoed kernel, equilibrium payment, transients)	full	L×4, S
background strain (threshold, log bound, absorption, sheath)	full	L×2, S
the budget $\int \gamma \leq KT^{1/6} + K_2 + K_3T$	full	L, S
circularization; the clock-ratio scoping	full	L×3, S, I
per-trajectory chain (transit factor, fold)	full	L×3, S
classification at the active scale	full	L, S
interval regularity; the BKM split	full	L, S
rescaling limit (Morrey identity, CKN floor, almost-periodicity)	full	L×2, S
confinement; derived defect	full	S
quantitative rigidity ($\theta = \frac{1}{2}$)	full	L×2, S
the squeeze ($\varepsilon < \varepsilon_7$)	full	L, S
the main assembly (Section 10)	full	—

APPENDIX C. INDEX OF MACHINE CERTIFICATES

The Lean project contains 54 kernel-checked theorems across seven files, zero unproven placeholders, over a pinned unmodified Mathlib; the build is reproduced by `lake exe cache get && lake build`. Abbreviated index (the repository’s master index carries each theorem’s adjudication log):

theorems	certify
<code>trace_cube_split</code> , <code>trace_SWW</code> , <code>trace_cube_det</code> , <code>financing_pointwise</code>	the financing identity: the trace split; the sign $\text{tr}(SW^2) = +\frac{1}{4}\omega^\top S\omega$; $\text{tr } S^3 = 3 \det S$; the assembly
<code>lemma18_pairing</code> , <code>lemma18_cauchy_schwarz</code> , <code>lemma18_org_bound</code>	the exact pairing $\alpha = \sum A_i \lambda_i$; its Cauchy–Schwarz step; the organization bound $\sqrt{2/3}$
<code>calibration_factor_closed_form</code> <code>qt1_scaling_coefficient</code> , <code>qt1_energy_identity</code>	the core-radius ruler factor $\frac{1}{2} - \frac{\ln 2}{3}$ exactly the localized defect energy identity and its exact $-\frac{1}{4}$ scaling balance

theorems	certify
moment_law_assembly, moment_law_equilibrium pancake_extremal_sq, veto_algebra, parity_kernel_odd, rayleigh_core	the second-moment law $4\nu - \alpha m_2$ and the Burgers equilibrium $m_2 = 4\nu/\alpha$ the discriminant inequality with pancake equality; the veto $S\hat{z} = 0$; the odd parity kernel; the Rayleigh sign core
w5_boundary_exponent, w5_window_slack, w6_closure, w3_lognorm_assembly	the adiabatic boundary exponent $\frac{1}{2}$; the window slack; the bootstrap root bound; the episode-rate assembly
w9_shell_series, w10_holder_exponents, w10_exponent_chain, w9_chebyshev w13_defect_gap	the cell-shell series $\frac{4}{3}$; the Hölder pairing $(\frac{5}{6}, \frac{1}{6})$; the exponent chain; the volume Chebyshev bound the dichotomy algebra of the superseded defect-gap argument (statement true; role demoted — see the record)
w7_duhamel_integral, w1_tail_integral, w8_tail_integral, gram_integral skew_core,	$\int_0^\infty e^{-as} = 1/a$; the kernel tails $r^{-3}/3$, $r^{-5}/5$; the Gram pairing $\langle r, W' \rangle = -2$ the $\mathbb{C}^2$ -skew content
hermitian_quadratic_real beta_cont, beta_deriv, beta_integrable, w16_beta_integral sc7_small_eps_choice	the Fujita–Kato beta integral $= \pi$, with its three supporting theorems the squeeze arithmetic: $\varepsilon < (\varepsilon_{\text{CKN}}/C_{\text{QT}})^{4/3} \Rightarrow C_{\text{QT}}\varepsilon^{3/4} < \varepsilon_{\text{CKN}}$
cp_veto_shifted_field, cp_axial_residue, cp_payment_algebra, cp_band_cancellation bg_log_absorption, bg_half_absorption ec_threshold, ec_bkm_split, ec_rate_floor	the shifted-field veto $S_{33} = 0$; the no- $\cos^2 \theta$ expansion; the equilibrium payment; the γ -free band sum the logarithmic absorption with structural coefficient $B/(A+B) < 1$; the sheath absorption the circularization bootstrap window; the exact split $(w-r)_+ + \min(w, r) = w$; the damping-rate floor
ft_transit_factor, ft_chain_fold, ft_middle_eigenvalue dc_coherence_inherit sc8_slice_scaling, sc8_holder_exponents su_green_integral, qd_exponent_positive, qd_rpow_collapse	the intensity-free transit factor θ^{*-1/c_0} ; the chain fold; the vetoed middle eigenvalue the downward coherence inheritance at $3\Delta_0$ the critical-slice scaling $r^2 \cdot r^{-3} = r^{-1}$; the L^2 – L^3 Hölder pairing the sector-uniformity kernel integral $1/(m+1)$; the depletion exponent $m - \frac{1}{2} \geq \frac{1}{2}$; its power collapse

Declared formalization targets (analysis-level; each proved in the analytic text): the torus integration by parts $\int \text{tr}(\nabla \mathbf{u})^3 = 0$; the moment-law and circularization ordinary-differential comparisons; interval-arithmetic eigenvalue enclosures for the swept normal form.

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Data availability. The Lean formalization (source and build files), the symbolic-adjudication scripts with their logs, and the instrument code with its archived outputs are available in the repository accompanying the paper, <https://github.com/truthintime/>

[navier-stokes-proof](#), public upon publication. The complete development record is available from the author upon request.

REFERENCES

- [Ash+87] W. T. Ashurst, A. R. Kerstein, R. M. Kerr, and C. H. Gibson. “Alignment of vorticity and scalar gradient with strain rate in simulated Navier–Stokes turbulence”. In: *Physics of Fluids* 30 (1987), pp. 2343–2353.
- [BCV19] J. Bedrossian, M. Coti Zelati, and V. Vicol. “Vortex axisymmetrization, inviscid damping, and vorticity depletion in the linearized 2D Euler equations”. In: *Annals of PDE* 5.1 (2019). Paper No. 4, 192 pp.
- [BKM84] J. T. Beale, T. Kato, and A. Majda. “Remarks on the breakdown of smooth solutions for the 3-D Euler equations”. In: *Communications in Mathematical Physics* 94 (1984), pp. 61–66.
- [CF93] P. Constantin and C. Fefferman. “Direction of vorticity and the problem of global regularity for the Navier–Stokes equations”. In: *Indiana University Mathematics Journal* 42 (1993), pp. 775–789.
- [CKN82] L. Caffarelli, R. Kohn, and L. Nirenberg. “Partial regularity of suitable weak solutions of the Navier–Stokes equations”. In: *Communications on Pure and Applied Mathematics* 35 (1982), pp. 771–831.
- [CL55] E. A. Coddington and N. Levinson. *Theory of Ordinary Differential Equations*. New York: McGraw–Hill, 1955.
- [CW17] D. Chae and J. Wolf. “Removing discretely self-similar singularities for the 3D Navier–Stokes equations”. In: *Communications in Partial Differential Equations* 42 (2017), pp. 1359–1374.
- [ESŠ03] L. Escauriaza, G. Seregin, and V. Šverák. “ $L_{3,\infty}$ -solutions of the Navier–Stokes equations and backward uniqueness”. In: *Russian Mathematical Surveys* 58 (2003), pp. 211–250.
- [Fef06] C. L. Fefferman. “Existence and smoothness of the Navier–Stokes equation”. In: *The Millennium Prize Problems*. Ed. by J. Carlson, A. Jaffe, and A. Wiles. Cambridge, MA: Clay Mathematics Institute and American Mathematical Society, 2006, pp. 57–67.
- [FK64] H. Fujita and T. Kato. “On the Navier–Stokes initial value problem. I”. In: *Archive for Rational Mechanics and Analysis* 16 (1964), pp. 269–315.
- [GS20] T. Gallay and D. Smets. “Spectral stability of inviscid columnar vortices”. In: *Analysis & PDE* 13 (2020), pp. 1777–1832.
- [Hop51] E. Hopf. “Über die Anfangswertaufgabe für die hydrodynamischen Grundgleichungen”. In: *Mathematische Nachrichten* 4 (1951), pp. 213–231.
- [Jim+93] J. Jiménez, A. A. Wray, P. G. Saffman, and R. S. Rogallo. “The structure of intense vorticity in isotropic turbulence”. In: *Journal of Fluid Mechanics* 255 (1993), pp. 65–90.
- [Kat84] T. Kato. “Strong L^p -solutions of the Navier–Stokes equation in $\mathbb{R}^m$, with applications to weak solutions”. In: *Mathematische Zeitschrift* 187 (1984), pp. 471–480.
- [Kid81] S. Kida. “Motion of an elliptic vortex in a uniform shear flow”. In: *Journal of the Physical Society of Japan* 50 (1981), pp. 3517–3520.

- [KS07] N. Kikuchi and G. Seregin. “Weak solutions to the Cauchy problem for the Navier–Stokes equations satisfying the local energy inequality”. In: *Nonlinear Equations and Spectral Theory*. Vol. 220. American Mathematical Society Translations, Series 2. Providence, RI: American Mathematical Society, 2007, pp. 141–164.
- [Lem02] P. G. Lemarié-Rieusset. *Recent Developments in the Navier–Stokes Problem*. Chapman & Hall/CRC Research Notes in Mathematics 431. Boca Raton, FL: Chapman & Hall/CRC, 2002.
- [Ler34] J. Leray. “Sur le mouvement d’un liquide visqueux emplissant l’espace”. In: *Acta Mathematica* 63 (1934), pp. 193–248.
- [Lin98] F. Lin. “A new proof of the Caffarelli–Kohn–Nirenberg theorem”. In: *Communications on Pure and Applied Mathematics* 51 (1998), pp. 241–257.
- [Mat95] P. Mattila. *Geometry of Sets and Measures in Euclidean Spaces: Fractals and Rectifiability*. Cambridge Studies in Advanced Mathematics 44. Cambridge University Press, 1995.
- [MH94] M. V. Melander and F. Hussain. “Core dynamics on a vortex column”. In: *Fluid Dynamics Research* 13 (1994), pp. 1–37.
- [MKO94] H. K. Moffatt, S. Kida, and K. Ohkitani. “Stretched vortices—the sinews of turbulence; large-Reynolds-number asymptotics”. In: *Journal of Fluid Mechanics* 259 (1994), pp. 241–264.
- [MS71] D. W. Moore and P. G. Saffman. “Structure of a line vortex in an imposed strain”. In: *Aircraft Wake Turbulence and Its Detection*. Ed. by J. H. Olsen, A. Goldberg, and M. Rogers. New York: Plenum Press, 1971, pp. 339–354.
- [NRŠ96] J. Nečas, M. Růžička, and V. Šverák. “On Leray’s self-similar solutions of the Navier–Stokes equations”. In: *Acta Mathematica* 176 (1996), pp. 283–294.
- [Olv74] F. W. J. Olver. *Asymptotics and Special Functions*. New York: Academic Press, 1974.
- [Sch+00] D. A. Schecter, D. H. E. Dubin, A. C. Cass, C. F. Driscoll, I. M. Lansky, and T. M. O’Neil. “Inviscid damping of asymmetries on a two-dimensional vortex”. In: *Physics of Fluids* 12 (2000), pp. 2397–2412.
- [Sch76] V. Scheffer. “Partial regularity of solutions to the Navier–Stokes equations”. In: *Pacific Journal of Mathematics* 66 (1976), pp. 535–552.
- [Tsa14] T.-P. Tsai. “Forward discretely self-similar solutions of the Navier–Stokes equations”. In: *Communications in Mathematical Physics* 328 (2014), pp. 29–44.
- [Tsa98] T.-P. Tsai. “On Leray’s self-similar solutions of the Navier–Stokes equations satisfying local energy estimates”. In: *Archive for Rational Mechanics and Analysis* 143 (1998), pp. 29–51.

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